Directed Reciprocity and Deep-Nesting Bounds for Matrix Transposition on Tapes

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October 2026, revised November 2026

Abstract

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $\max(1, \lceil \log_2 d \rceil)$-bit entries in time linear in the input length? By the fourth paper of this series, excluding oblivious linear-time transposers splits into two combinatorial problems: a storage-code deficit on the actual cone relations large enough to exceed the reveal cost (the coding crux), and cheap depth reduction of head-walk dependency graphs (the trajectory cut problem). Bounded degree and paired-distance separation alone do not imply half rate [11]. This paper advances three frontiers left open by the seventh, eighth and ninth papers of the series, and every new numbered theorem, proposition, lemma and corollary is kernel-verified in Lean 4.

First, for nested elimination on directed recovery relations, we give a purely positional characterization of the nestedness hypothesis of the seventh paper, show that every reciprocal pair forces distinct record lists, and prove a sharp dichotomy for chordless digon cycles with at most one active outside parent per vertex: a piece certificate exists over some number of records if and only if the outside arcs do not form a unidirectional cycle of stride 3 (Theorem DC). In particular, the 15-vertex obstruction X_{15} has no piece certificate on its unique minimum nonempty core over any number of records, and the Petersen graph $\mathrm{GP}(5, 2)$ has no global elimination certificate over any number of records, whereas $\mathrm{GP}(7, 2)$ and $\mathrm{GP}(9, 2)$ require minimum record counts 4 and 3, respectively (Proposition GP), resolving two open formalization problems of the seventh paper. Using minimal chordal digon paths and hub-augmented stride-3 pieces, we then prove that the entire class $\mathcal{S}$ of Theorem S of the fourth paper—transitive-tournament-free relations of maximum in-degree 3 in which every vertex of in-degree 3 has only reciprocal children—admits a certified piece decomposition and satisfies the half-rate bound $|C|^2 \leq |\alpha|^{|V|}$ over every finite alphabet and for all codes, linear or nonlinear (Theorem S^*), resolving the all-code Theorem S target of the seventh paper, together with a directed barrier inequality (Corollary B^*) and a 21-vertex strongly connected triangle-free witness $\mathrm{G}_{21} \in \mathcal{S}$ on which earlier peeling rules stall (Proposition G21).

Second, for scalar linear codes on triangle-free graphs of maximum degree 4 defined as the kernel of a matrix M with unit diagonal over an arbitrary field, we remove the heavy-separation hypothesis (S2) of the eighth paper's Theorem SH: adjacent degree-4 exceptional vertices and arbitrary matchings of exceptional vertices are allowed, provided no two exceptional vertices share a common neighbor (Theorem SH^*), yielding a matching bound controlled by the heavy-matching invariant $\rho_{\mathrm{m}}(G, H) \geq \rho_3(G, H)$ (Corollary M^*).

Third, on the deep-nesting walk $U(s_0, m, 1)$, we formalize the recursive stroke structure and $O(m)$ turn-potential evaluator (Proposition SL), prove an m-free mod-q straddle lower bound and exact parity-conditioned straddle count $\Pi_\kappa^2/4$ (Theorem V^*), establish a super-Fibonacci unwrapping theorem showing that level-uniform single-head cancellation admits at most $2\lfloor \log_2 q \rfloor + 2$ consecutive levels whose blind apex and band-$0, 1, 2$ errors are at most $q/8$ (Theorem LU), determine the block-cancellation drift and Mersenne $q = 2^B - 1$ binary-clock failure residue (Proposition OC), and prove an unconditional three-regime witness theorem with zero $\log q$ loss for block-label pairs of bounded block length across every $K_1, K_2 \in \mathbb{Z}$ and

every modulus $q \geq 3$ at two fixed odd shifts, or a single shift for symmetric pairs (Theorem A; translating its single-head regimes to actual two-head hop errors requires blindness at the chosen shift). The linear-time transposition question itself remains open.

Contributions. Hans Brende chose the problem, fixed the verification standard, and directed the research. Gemini 4 Argon, a large language model developed by Google DeepMind, developed the mathematics, wrote the Lean formalization, and wrote this paper under Hans Brende's direction. Section 3 builds on the nested elimination engine and piece-decomposition framework of *Nested Elimination and Half-Rate Bounds for Storage Codes* by Hans Brende and Claude Opus 5.5 (Anthropic) [10] and on the class of directed recovery relations introduced in *Curved Motion, Depth Reduction, and Storage Codes for Matrix Transposition on Tapes* by the same authors [8]. Sections 4 and 5 extend the minimal-counterexample method for degree-4 scalar linear codes and the deep-nesting walk and block-label analysis of *Bounded Nesting and Depth Reduction for Matrix Transposition on Tapes*, also by Hans Brende and Claude Opus 5.5 (Anthropic) [7], and situate these results alongside the source-cover barriers of *Source Covers and Barriers to Tape Transposition* by Hans Brende and GPT-6 Astra (OpenAI) [14].

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1 Introduction

Does there exist a deterministic multitape Turing machine that transposes every $d \times d$ matrix of $b(d)$-bit entries, where $b(d) = \max(1, \lceil \log_2 d \rceil)$ and $L = d^2 b(d)$ is the bit-length of the input, in time $O(L)$? Following the earlier papers of this series [9], we write (T) for this assertion:

> (T) There exists a deterministic multitape Turing machine—with a binary read-only two-way input tape, a binary write-only one-way output tape, and a fixed finite number of two-way read/write work tapes over a fixed finite work alphabet—that transposes $d \times d$ matrices of $b(d)$-bit entries in $O(d^2 b(d))$ steps.

By a theorem of Harvey and van der Hoeven [19], N-bit integer multiplication reduces in linear time on a multitape Turing machine to matrix transposition; hence refuting (T) would establish an $\Omega(N \log N)$ lower bound for deterministic off-line multitape integer multiplication. Whether (T) holds remains open.

1.1 Context in the series

The fourth paper of this series [8] reduced the impossibility of *oblivious* linear-time transposers to the conjunction of a depth-reduction statement and a coding bound on the resulting side information:

(i) *The trajectory cut problem* (TCP(k)): in every k-head walk dependency DAG of length T, deleting at most εT edges reduces the longest directed path to length $O_{\varepsilon,k}(1)$.

(ii) *The coding crux*: storage codes on the actual cone relations have a rate deficit large enough to exceed the reveal and trimming costs at the depth supplied by the first statement. A half-rate bound would suffice with sufficiently cheap revelation. An abstract half bound under bounded degree and paired-distance separation is false [11]; a positive degree-dependent deficit alone is elementary and need not beat those costs. Conjecture P of [10] instead asks for half rate on loopless transitive-tournament-free (TT-free) relations of in-degree at most 3, and remains open.

Subsequent papers sharpened both halves of this program while exposing concrete obstructions to naive extensions:

- On the *directed coding* side, the fourth paper [8] proved the half-rate bound for *scalar linear* codes on the mixed class $\mathcal{S}$ of TT-free relations with maximum in-degree 3 in which every vertex of in-degree 3 has only reciprocal children (its Theorem S). The seventh paper [10] introduced the nested elimination engine over general polymatroids (removing linearity altogether), proved the all-code half-rate bound on undirected subcubic triangle-free graphs (its Theorem N3) and on loopless TT-free relations with subcubic underlying graph (its Theorem U3), and constructed the 15-vertex TT-free orientation X_{15} of maximum in-degree 3 on which minimum-core peeling fails. Within item 2 of [10, Section 9.2], alongside Conjecture P ([10, Conjecture 9.1], for general in-degree at most 3), it singled out as an intermediate target whether the fourth paper's Theorem S lifts from scalar linear codes to *all* codes on the entire class $\mathcal{S}$, and noted two unformalized impossibility claims [10, Remarks 7.3 and 7.5; Section 9.2, item 6]: that X_{15} admits no piece certificate on its unique minimum nonempty core over any number of records, and that the Petersen graph admits no global elimination certificate over any number of records.

- On the *degree-4 linear coding* side, scalar binary storage codes reach rate 5/8 on the degree-5 Clebsch graph (Lean-verified in [8, Corollary 6.11]), and the seventh paper's degree-4 triangle-free example has an $\mathbb{F}_4$-linear storage code of rate $7/13 > 1/2$ [10, Example 7.8]. Its odd girth is five; half rate at degree four with growing odd girth remains open. Tree-local arguments cannot handle arbitrary clusters of degree-4 vertices [8, Proposition 6.28]. The eighth paper [7] isolated the exceptional set H of degree-4 rows in a unit-diagonal check matrix $M = I + B$ on a triangle-free graph of maximum degree 4 and proved $\dim\ker M \le |V|/2$ over every field under two hypotheses on H: (S1) no two vertices of H share a common neighbor, and (S2) no two vertices of H are adjacent. Whether (S2) is genuinely needed or an artifact of the induction was left open [7, Section 8.2, Remark 8.8; Section 10, Problem 10.4].

- On the *head-motion and deep-nesting* side, the eighth paper [7] proved TCP(k) for all schedules of bounded nesting load (its Theorem F) and isolated the 2-head *deep-nesting pair* DNP(s_0, m), running two shifted copies of the recursive nested walk $U(s_0, m, 1)$, as the sharpest open case of TCP(2). On a single copy of $U(s_0, m, 1)$, the eighth paper's turn counter [7, Section 5.2, Theorem 5.2 and Corollary 5.5] (built on Lemma Zq of [8, Section 5.5, Lemma 5.15]) cuts depth to $2^{m+O(1)}/\varepsilon$; the eighth paper introduced block-cancellation labels $B(a, K)$ to probe whether two heads can cancel turn-counter drift across scales, stated Conjecture A″ [7, Section 5.6, Conjecture 5.24]—that at some odd shift, every mod-q label of cost at most $n_{\mathrm{h}}/8$ on the r-fold cover has at most $C \log_2 q + C' + \eta(m+1)$ doubly θ_0-served levels ($\eta < 1$)—and at the end of [7, Section 5.6] identified two open steps for proving Conjecture A″ even on block-label pairs: (1) *"a bound on $|F|$ on the straddling hops, or a version of Theorem V that counts the window directly"*, and (2) *"at one fixed shift for all coarse classes, equidistribution of a class's apex centres relative to the other head's canceller valleys of level about $k + \log_2(m-k)$, a mixed-radix carry problem"*. Meanwhile, the ninth paper [14] developed an independent bypass of TCP via complete source covers and return-profile barriers on restricted return geometries, leaving open the unrestricted adaptive regime.

1.2 Summary of main results

This paper resolves the intermediate directed coding target of [10, Section 9.2, item 2] (while Conjecture P, its Conjecture 9.1, remains open), removes hypothesis (S2) from Theorem SH of [7], and proves and formalizes structural stroke, straddle, and super-Fibonacci laws together with three-regime failure witnesses for bounded-range block-label pairs on the deep-nesting walk. Every new numbered result in the paper is verified in Lean 4 using only the three standard foundational axioms (Section 2.5 and Appendix B).

1. Directed reciprocity, digon-cycle dichotomy, and the all-code Theorem S* (Section 3). We begin by recasting the stateful recursive predicate `HalfRate.RelCharged` of [10] in purely positional terms (Proposition 3.1): an elimination list $l = [v_0, \dots, v_{k-1}]$ with records $\{P_r\}_{r \in I}$, supplying-record map b, and optional outside-parent map par is relationally nested from empty initial state if and only if every step covers its active parents from the revealed prefix S_j, its supplying record $P_{b(v_j)}$, and its optional outside parent, and whenever $j < j'$ share the same outside parent w, $P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$ (for disjoint records, either $b(v_j) = b(v_{j'})$ or all of $P_{b(v_j)}$ is revealed before step j'). As an immediate invariant on any piece certificate, no two vertices joined by a reciprocal pair (a *digon*) can belong to the same own record $P_{a(u)}$, so every connected digon component on at least two vertices requires $m \ge 2$ records.

We then classify piece certificates on chordless digon cycles C_L ($L \geq 4$) in TT-free relations where each cycle vertex has at most one active parent outside $U \cup C_L$ (Proposition 3.2 and Theorem 3.4): an m-record piece certificate exists on C_L for *some* $m \geq 1$ if and only if the outside parents do not form a *stride*-3 *hub triple*, that is, $3 \mid L$ together with outside vertices $w_0, w_1, w_2 \notin U \cup C_L$ such that $w_{i \bmod 3}$ is an active parent of cycle vertex i for every $0 \leq i < L$. When the stride-3 hub obstruction is absent, $m = 2$ records suffice for $L = 4$ and $m = L + 1$ records suffice via two-sided arc expansion for every $L \geq 5$ (while on pure stride-s hub cores C_{sk}, $m = 2$ suffices for even $s \geq 4$ and $m = 3$ suffices for every $s \geq 4$); when the stride-3 hub triple is present, a positional locking invariant rules out piece certificates across *all* $m \geq 1$. Specializing to the unique minimum nonempty core $\{0, \ldots, 5\}$ of X_{15} (Corollary 3.3) and to generalized Petersen graphs (Proposition 3.5) resolves both parts of [10, Remarks 7.3 and 7.5; Section 9.2, Problem 6] in Lean 4 and reveals a non-monotone global-certificate record hierarchy: $\mathrm{GP}(5, 2)$ (the Petersen graph) requires ∞ records globally, $\mathrm{GP}(7, 2)$ requires exactly 4 records, and $\mathrm{GP}(9, 2)$ drops back to 3 records.

Using these cycle tools, we prove **Theorem S*** (Theorem 3.6): every relation N in the full class $\mathcal{S}$ of Theorem S of [8] admits a certified piece decomposition (`HalfRate.Decomp`), so every valid polymatroid h on V obeys $2h(V) \leq \sum_{v \in V} h(\{v\})$ and every storage code $C \subseteq \alpha^V$ (linear or nonlinear, over any finite alphabet α) satisfies $|C|^2 \leq |\alpha|^{|V|}$. The key mechanism overcoming embedded stride-3 hub cycles inside an in-closed strongly connected component S of $\mathcal{S}$ is *hub augmentation* (`exists_piece_of_scc_classS`): if S has no minimal chordal digon arc and a shortest digon cycle $C_L \subseteq S$ is obstructed as a standalone piece, then C_L is a stride-3 hub cycle with hubs $w_0, w_1, w_2 \in S \setminus C_L$; strong connectivity forces an outgoing arc from some $u_{i_0} \in C_L$ to $S \setminus C_L$, which $\mathcal{S}$ reciprocates into a digon between u_{i_0} and its hub w_{r_0} while girth minimality forces w_{r_0} to have in-degree 2 and no other child on C_L, so adjoining w_{r_0} yields an $(L + 2)$-record sequential nested piece certificate on $C_L \cup \{w_{r_0}\}$. We deduce a directed vertex-cut and isolated-outside-vertex barrier inequality for all $\mathcal{S}$ relations (Corollary 3.7) and construct an explicit 21-vertex strongly connected triangle-free witness $\mathrm{G}_{21} \in \mathcal{S}$ (Proposition 3.8) on which low-in-degree peeling, one-way cycle peeling, and shortest-digon-cycle rules all fail simultaneously, while Theorem S* certifies rate at most $1/2$.

2. Scalar linear codes at degree 4 without heavy separation (Section 4). In Section 4 we prove **Theorem SH*** (Theorem 4.1): over any field F, if $M \in F^{V \times V}$ has unit diagonal, no recovery triangle, and satisfies ($\star 4$) at every vertex (for instance when M is supported in a triangle-free simple graph $G = (V, E)$ of maximum degree 4), and the heavy set H_M of vertices violating ($\star\star$) either has size at most 2 or satisfies (S1) alone—no vertex has two distinct heavy recovery neighbors (in graph form, no two distinct vertices of H_M share a common neighbor in G)—then $\dim_F \ker M \leq \lfloor |V|/2 \rfloor$ (and likewise under the same hypotheses on M^{T}). Hypothesis (S2) of [7] is eliminated completely: $G[H_M]$ may contain arbitrary induced matchings of adjacent degree-4 vertices. The proof replaces Phase E of [7] with a unified final count (`F3_star`): under (S1) alone, a pending triple of a heavy vertex h contains at most one heavy member g, leaving two light members u_1, u_2 whose shared pending pair $\{a, b\}$ and neutral dimension drop contradict linear independence uniformly for all $|W \setminus X_0| \geq 3$, while in minimal counterexamples any degree-4 heavy vertex still retains at least three light neighbors under (S1). Combining Theorem SH* with Tutte–Berge deficiency yields **Corollary M*** (Corollary 4.2), improving the eighth paper's distance-3 packing invariant $\rho_3(G, H_M)$ to the common-neighbor-free packing invariant $\rho_{\mathrm{m}}(G, H_M) \geq \rho_3(G, H_M)$.

3. Deep-nesting stroke laws, super-Fibonacci unwrapping, and bounded-range block pairs (Section 5). In Section 5 we formalize the algebraic and combinatorial theory of turn-

supported periodic labels modulo q on the deep-nesting walk $U(s_0, m, 1)$ and on the deep-nesting pair $\mathrm{DNP}(s_0, m)$ (thereby also providing Lean 4 counterparts for the unformalized stroke, band, block and straddle lemmas of [7, Section 5]):

- **Proposition SL** (Proposition 5.1) establishes the exact stroke decomposition bijection in $[0, \Pi_0)$, proves that under fair binary branch choices the stopping depth before the nested walk visits m has distribution $2^{-(s+1)}$ on $\{0, \dots, m-1\}$ independently of directional parity, and constructs an $O(m)$ recursive turn-potential evaluator `evalPhiRec` proved equal to cumulative turn jumps over a full period $P = 4\Pi_0$.

- **Theorem V*** (Theorem 5.2) proves an m-free mod-q straddle involution bound: whenever a level-k apex hop of one head straddles a level-κ V-turn of the other head ($\Pi_\kappa \le R_k$), the paired half-widths $(d, \Pi_\kappa - d)$ cancel the nested potential identically over $\mathbb{Z}$, forcing their reduced mod-q errors to sum to at least $(2A' + 2(c_{\mathrm{f}}(\kappa) - c_{\mathrm{b}}(\kappa))) \bmod q$; moreover, summing the straddle set size over centers of either fixed parity across a $2\Pi_\kappa$-window yields the *exact* triangular identity $(\Pi_\kappa/2)^2 = \Pi_\kappa^2/4$.

- **Theorem LU** (Theorem 5.3) proves that for any level-uniform turn-supported label modulo $q \ge 3$, if $s \ge 1$ consecutive levels have single-head reduced apex and valley band-$0, 1, 2$ errors at most $q/8$, the modular band relation unwraps over $\mathbb{Z}$ with zero wrap-around and forces the super-Fibonacci inequality $b_0(\ell) \ge b_0(\ell+1) + b_0(\ell+2) + 4$, whence $32(F_s - 1) \le q$ and $s \le 2\lfloor \log_2 q \rfloor + 2$.

- **Proposition OC** (Proposition 5.4) proves exact forward and backward block-cancellation bounds for $B(a, K)$, establishes strictly positive mean drift $+(a(K+1)+1)/2$ per $(a+1)$-level block across all 2^{a+1} stroke-direction choices, and shows that the pure binary clock $B(\infty, 0)$ fails modulo every Mersenne number $q = 2^B - 1$ ($B \ge 2$) with straddle residue $q - 1$ at every third/periodic scale $B \mid (m - \kappa + 2)$.

- **Theorem A** (Theorem 5.5) combines these ingredients for block-label pairs $(B(a_1, K_1), B(a_2, K_2))$ of bounded block length ($1 \le a_1, a_2$): Theorem V* completely bypasses open step (1) of [7, Section 5.6] for level-uniform labels (paired straddle sums need no bound on $|F|$ and no window assumption on q), while `exists_oddShift_coarse_straddle` establishes a one-direction $\frac{1}{16}$-fraction form of step (2) (weaker than the all-coarse-classes hypothesis of [7, Proposition 5.23], which is obstructed by prefix lockstep of nested V-apices, `apexOffset_diff_indep_prefix`). Whenever $m < s_0$ and $4(\lfloor \log_2(m+1) \rfloor + 1 + \max(a_1, a_2)) \le m+1$, Theorem A produces two fixed odd shifts ξ_1, ξ_2 (a single odd shift ξ for symmetric pairs $a_1 = a_2, K_1 = K_2$) giving three-regime failure witnesses on $\Omega(m)$ levels across *every* $K_1, K_2 \in \mathbb{Z}$ and *every* modulus $q \ge 3$, with *zero* $\log q$ loss: blind single-head error witnesses in Regimes 1 and 2, and two-head coarse straddle witnesses in Regime 3. We also record the four remaining steps separating these witnesses from Conjecture A$''$ (Remark 5.6).

1.3 Related work

The equivalence between locally recoverable storage capacity on a digraph G, index coding broadcast rate on the complement, and the guessing number $\mathrm{gn}(G) = \sup_{|\alpha| \ge 2} \sup_C \log_{|\alpha|} |C|$ (over locally recoverable codes $C \subseteq \alpha^V$, [22, 10]) goes back to Birk and Kol [6], Riis [22], Bar-Yossef, Birk, Jayram and Kol [2], Gadouleau and Riis [17], and Mazumdar [20]. On undirected triangle-free graphs, Christofides and Markström [16] bounded the guessing number by the fractional matching number (giving rate $1/2$ for all undirected triangle-free graphs only on symmetric recovery neighborhoods),

Cameron, Dang and Riis [15] classified high-guessing triangle-free graphs in low degree, Atkins, Rombach and Skerman [1] determined guessing numbers of odd cycles, and Mazumdar, McGregor and Vorotnikova [21] proved half-rate bounds under sparse global average degree, whereas without a degree bound Barg and Zémor [4] and Barg, Schwartz and Yohananov [3] constructed coset and Cayley storage codes on triangle-free graphs of growing degree with rates exceeding 1/2 (approaching 3/4 and 1, respectively). On directed graphs without undirected symmetry, undirected fractional-matching duality breaks down because of directed cycles of pure one-way arcs; Theorem S* (Theorem 3.6) is, to our knowledge, the first all-code half-rate theorem covering a class of TT-free digraphs that simultaneously permits full subcubic undirected reciprocity, unbounded directed one-way cycles, and mixed components combining both.

On the complexity side, reductions from network coding and circuit depth to Valiant's edge-removal problem go back to Valiant [23]; see [9, 8, 7, 14] for a detailed account of the connections between TCP(k), monotone and non-monotone depth reduction, matrix rigidity, and source covers.

1.4 Organization and formal reading guide

Section 2 fixes notation from the earlier papers of the series and explains our Lean 4 verification conventions (Section 2.5). Section 3 develops positional nestedness, the digon-cycle dichotomy, the generalized Petersen record hierarchy, and Theorem S*. Section 4 proves Theorem SH* and Corollary M*. Section 5 proves Proposition SL, Theorem V*, Theorem LU, Proposition OC, and Theorem A on the deep-nesting walk and pair. Section 6 synthesizes what any hypothetical linear-time transposer must look like in light of these bounds, and Section 7 collects the sharpened open problems. Appendix A indexes every Lean declaration cited in the paper, and Appendix B gives exact instructions for building the library, checking axioms, and reproducing the finite numerical examples.

2 Preliminaries

We keep the notation and conventions of the fourth, seventh and eighth papers [8, 10, 7] so that cited declarations align directly with their Lean 4 libraries. Throughout, logarithms are base 2, $\mathbb{N} = \{0, 1, 2, \dots\}$, and for integers $a \le b$ we write $[a, b] = \{x \in \mathbb{Z} : a \le x \le b\}$ and $[a, b) = \{x \in \mathbb{Z} : a \le x < b\}$ (or their subsets in $\mathbb{N}$ when the endpoints are natural numbers). For $x \in \mathbb{Z}$ and $q > 0$, we write $x \bmod q \in \{0, 1, \dots, q-1\}$ for the non-negative Euclidean remainder (`Int.emod` in Lean).

2.1 Turing machines, transposition, and the depth–coding split

In the machine model of [9, 8] (`Transpose.TM`), a deterministic multitape Turing machine has a binary read-only two-way input tape, a binary write-only one-way output tape, and k two-way read/write work tapes over a finite work alphabet $\{0, 1, \dots, g\}$. Given a $d \times d$ matrix of $b(d)$-bit entries in row-major order, where $b(d) = \max(1, \lceil \log_2 d \rceil)$ and $L = d^2 b(d)$, the machine must halt with the transpose in row-major order on the output tape; statement (T) is formalized as `Transpose.LinearTimeTransposable` (see [9, Section 2], Lean-verified). A transposer is *oblivious* if its head trajectories, output-append times, and halting time depend only on the input dimension d [9, Section 2]. By Theorem D of [8, Section 4.2, Theorem 4.10] (proved in the text of [8]; not formalized in Lean), if the k-head walk dependency DAG of an oblivious linear-time transposer admits εT-edge cuts to bounded hop-depth (TCP(k)) and storage codes on the resulting bounded-in-degree TT-free recovery relations have rate bounded away from 1, then no such transposer exists.

2.2 Directed recovery relations, polymatroids, and nested elimination

Let V be a finite vertex set. A *recovery relation* on V is a parent map $N : V \to \mathcal{P}(V)$; we write $u \to v$ when $u \in N(v)$ (u is a *parent* of v, and v is a *child* of u). The relation N is *irreflexive* (loopless) if $v \notin N(v)$ for all $v \in V$. Two distinct vertices u, v form a *reciprocal pair* or *digon* if $u \in N(v)$ and $v \in N(u)$, and $u \to v$ is a *pure one-way arc* if $u \in N(v)$ and $v \notin N(u)$. We say N is *transitive-tournament-free* (`HalfRate.TTFree`) if there are no three distinct vertices $a, b, c \in V$ with $a \in N(b)$, $b \in N(c)$ and $a \in N(c)$; notice that an irreflexive symmetric relation is TT-free if and only if its underlying undirected simple graph is triangle-free. For a subset $S \subseteq V$, the induced restriction $N|_S$ on S sets $N|_S(v) = N(v) \cap S$ (formalized on the subtype S as `DirectedRecovery.inducedNeighborhood` in [12], and on V with empty parent sets outside S as `resRel` in Section 3.3).

A *storage code* on (V, N) over a finite nonempty alphabet α (`MotionDepthCodes.IsStorageCode`) is a nonempty subset $C \subseteq \alpha^V$ such that for every $v \in V$, the coordinate x_v of $x \in C$ is determined by its parent coordinates $(x_u)_{u \in N(v)}$. Sampling $X \in C$ uniformly defines a Shannon polymatroid $h : \mathcal{P}(V) \to \mathbb{R}$ (`MotionDepthCodes.IsPolymatroid`) satisfying $h(\emptyset) = 0$, monotonicity, submodularity, and validity $h(N(v) \cup \{v\}) = h(N(v))$ at every $v \in V$ (`MotionDepthCodes.IsValid`; see [8, Section 4] and [10, Section 2], Lean-verified). We say (V, N) satisfies the polymatroid half-rate bound (`MotionDepthCodes.RateBound` at $1/2$) if every polymatroid h valid for N obeys

$$2h(V) \le \sum_{v \in V} h(\{v\}), \tag{1}$$

which specializes on any storage code $C \subseteq \alpha^V$ to $2 \log |C| \le |V| \log |\alpha|$, i.e., $|C|^2 \le |\alpha|^{|V|}$ (`MotionDepthCodes.card_sq_le_of_rateBound_half`).

We recall the multi-record nested elimination engine of [10, Section 3]. Let $U \subseteq V$ be a *context* (already contracted vertices), $H \subseteq V \setminus U$ be a *piece*, and $W = V \setminus (U \cup H)$ be the set of *outside* vertices. *Elimination data* for H over an index set I of $m = |I|$ records consist of subsets $P_r \subseteq H$ ($r \in I$), an elimination list $l = [v_0, \dots, v_{k-1}]$ with underlying set H, an *own record* map $a : V \to I$ and *supplying record* map $b : V \to I$ with $v \in P_{a(v)}$ and $b(v) \neq a(v)$ for all $v \in l$, and an optional outside-parent map $\mathrm{par} : V \to \mathrm{Option}\ V$. At step $j < k$, given an initial revealed set S_0, write $S_j = S_0 \cup \{v_0, \dots, v_{j-1}\}$ (`HalfRate.rev`) for the revealed prefix and $Q_j = S_j \cup P_{b(v_j)}$ for the supply. In [10], relational nestedness (`HalfRate.RelCharged` applied to $N, U, P, b, \mathrm{par}, W, S_0, L_0, l$) is defined recursively along l by threading a state map $L : V \to \mathcal{P}(V)$ recording for each outside vertex $w \in W$ the set $Q_{j'} \cup \{v_{j'}\}$ at the most recent step j' charged to w:

- if $\mathrm{par}(v_j) = \mathrm{none}$ (*free step*), $N(v_j) \subseteq U \cup Q_j$, and L is unchanged;
- if $\mathrm{par}(v_j) = \mathrm{some}\ w$ (*charged step*), $w \in W$, $N(v_j) \subseteq U \cup \{w\} \cup Q_j$, and $L(w) \subseteq Q_j$, after which $L(w)$ is updated to $Q_j \cup \{v_j\}$.

By Lemma E of [10, Section 3.1, Lemma 3.2] (Lean-verified as `HalfRate.nested_elimination`), whenever g is a polymatroid and the charging is nested, $2g(H) \le \sum_{r \in I} g(P_r) + \sum_{w \in W} I_g(w; H)$. A *piece certificate* for H in context U (`HalfRate.PieceCert`) bundles $m \in \mathbb{N}$ pairwise disjoint records $P_0, \dots, P_{m-1}$ partitioning H (note $a(v) \neq b(v)$ forces $m \ge 2$ whenever $H \neq \emptyset$), a list l with $l.\mathrm{toFinset} = H$, maps $a, b : V \to \{0, \dots, m-1\}$ with $v \in P_{a(v)}$ and $a(v) \neq b(v)$ for $v \in l$, and $\mathrm{par} : V \to \mathrm{Option}\ V$ satisfying `HalfRate.RelCharged` from initial state $S_0 = \emptyset$, $L_0 \equiv \emptyset$ with outside set $W = V \setminus (U \cup H)$. A *global elimination certificate* over m records (`HalfRate.ElimCert`) is the global special case $U = \emptyset$, $H = V$ (where $W = \emptyset$ and every step is uncharged). A *decomposition* of N from context U (`HalfRate.Decomp`) is a sequence of disjoint pieces $H_1, \dots, H_t$ partitioning

$V \setminus U$ such that each H_s has a piece certificate in context $U \cup H_1 \cup \cdots \cup H_{s-1}$. By Theorem D of [10, Section 3.2] (Lean-verified as `HalfRate.rateBound_of_decomp` and `HalfRate.card_sq_le_of_decomp`), a decomposition from context $\emptyset$ implies (1) and $|C|^2 \le |\alpha|^{|V|}$ for every storage code $C \subseteq \alpha^V$.

2.3 Scalar linear codes on triangle-free graphs of maximum degree 4

Let $G = (V, E)$ be a finite simple undirected graph and F be an arbitrary field. Following [7, Section 8], a matrix $M \in F^{V \times V}$ has unit diagonal and support in G if $M_{v,v} = 1$ for all $v \in V$ and $M_{v,u} \neq 0 \implies \{v, u\} \in E$ for $u \neq v$. Its subspace $\ker M = \{x \in F^V : Mx = 0\}$ is a scalar linear storage code on G: each coordinate $x_v = -\sum_{u \neq v} M_{v,u} x_u$ is recovered linearly from its active recovery neighbors $N_M(v) = \{u \in V \setminus \{v\} : M_{v,u} \neq 0\}$ (`MotionDepthCodes.recSet`). Following [8, Section 6.6] and [7, Section 8.2], a vertex v satisfies ($\star\star$) (`MotionDepthCodes.Star`) if either $|N_M(v)| \le 2$, or $|N_M(v)| = 3$ and v reciprocates every outgoing recovery arc ($v \in N_M(u) \implies u \in N_M(v)$); it satisfies ($\star 4$) (`BoundedNesting.Star4`) if either $|N_M(v)| \le 3$, or $|N_M(v)| = 4$ and v reciprocates every outgoing recovery arc (automatically satisfied whenever M is supported in an undirected graph G of maximum degree at most 4). We call $v \in V$ *heavy* (or exceptional) if v violates ($\star\star$), and write $H_M \subseteq V$ (`BoundedNesting.heavyVerts`) for the set of heavy vertices. Theorem SH of [7, Section 8.2, Theorem 8.3] (Lean-verified as `BoundedNesting.theoremSH`) proved $\dim_F \ker M \le \lfloor |V|/2 \rfloor$ assuming ($\star 4$) everywhere together with $|H_M| \le 2$ or *both* (S1) no vertex has two distinct heavy recovery neighbors (`BoundedNesting.SepS1`), and (S2) no heavy vertex has a heavy recovery neighbor (`BoundedNesting.SepS2`).

2.4 Walks, turn labels, and the deep-nesting walk $U(s_0, m, 1)$

A 1-head walk on $\mathbb{Z}$ is a map $x : \mathbb{N} \to \mathbb{Z}$ with $|x(t+1) - x(t)| \le 1$; for k work heads, each head $h \in \{0, \ldots, k-1\}$ moves by $\{-1, 0, +1\}$ at every step. Following [7, Section 2.3] (`BoundedNesting/WalkDag.lean`), the *walk dependency DAG* over horizon T has vertex set $\{0, \ldots, T-1\}$, with *state edges* $t \to t-1$ for $1 \le t < T$ and *long-hop edges* $u \to p$ whenever some head visits the same cell at times p and u with $p + 2 \le u < T$ and no intermediate visit in (p, u) (`BoundedNesting.allHops`). By the label-cut principle [7, Section 2.3, Lemma 2.3] (Lean-verified as `BoundedNesting.labelCut_depth_walkDag`), for any label $\Psi : \mathbb{N} \to \{0, \ldots, q-1\}$ ($q \ge 1$), deleting the state edges at step descents $\Psi(t) < \Psi(t-1)$ (`BoundedNesting.descentsOf`) and the violated hop edges with $\Psi(u) \le \Psi(p)$ (`BoundedNesting.violatedOf`) leaves at most $q - 1$ long-hop edges along any directed path in the walk DAG. On a cyclic schedule whose long hops partition into closed cell chains, Lemma Zq of [8, Section 5.5, Lemma 5.15] (Lean-verified as `MotionDepthCodes.lemma_Zq` and specialized to turn-counting labels in `BoundedNesting.turnCounter_cost`) expresses q times the total number of cut descents and violated hops identically as the sum of step increments $(\Psi(t) - \Psi(t-1)) \bmod q$ plus reduced hop errors $(\Psi(u) - \Psi(p) - 1) \bmod q$.

For integers $0 \le m \le s_0$, define the dyadic scales $R_i = 2^{s_0 - i}$ for $0 \le i \le s_0$ (`scaleR`) and the nested *stroke lengths* $\Pi_\kappa = (m - \kappa + 1) R_\kappa$ for $0 \le \kappa \le m$ (`strokeLen`). As in [8, Definition 5.3] and [7, Section 2.5], the *deep-nesting walk* $U(s_0, m, 1)$ has fundamental period $P = 4\Pi_0 = 4(m+1)2^{s_0}$ (`dnpPeriod`) consisting of four level-0 strokes (forward, backward, backward, forward). Recursively, for $\kappa < m$, a level-κ stroke of duration $\Pi_\kappa = R_\kappa + 2\Pi_{\kappa+1}$ executes a plain segment of R_κ unit steps, followed by a level-$(\kappa+1)$ *back stroke* of duration $\Pi_{\kappa+1}$ reversing direction across those R_κ cells (making an *apex* turn at offset R_κ), followed by a level-$(\kappa+1)$ *forward stroke* of duration $\Pi_{\kappa+1}$ restoring the original direction (making a *valley* turn at offset $R_\kappa + \Pi_{\kappa+1}$); at the base level $\kappa = m$, a stroke is simply $R_m = 2^{s_0 - m}$ plain steps. On a single copy of $U(s_0, m, 1)$ (and on $\mathrm{DNP}(s_0, m)$ at

any shift), the turn counter of [7, Section 5.2, Theorem 5.2 and Corollary 5.5] cuts hop-depth to $2^{m+O(1)}/\varepsilon$ at edge density ε. The *deep-nesting pair* $\mathrm{DNP}(s_0, m)$ runs head 1 on $U(s_0, m, 1)$ and head 2 on a time-shifted copy $t \mapsto U(s_0, m, 1)((t + \xi) \bmod P)$ over a common tape.

2.5 Formal status of the results

Every numbered theorem, proposition, lemma, and corollary introduced in Sections 3 to 5 has a complete, kernel-checked proof in the accompanying Lean 4 library `lean/DirectedReciprocity/` (Lean and Mathlib version `v4.34.1`). Immediately below each numbered statement, a header line

Lean: `declaration_1`, `declaration_2` [`Module.lean`]

names the exact Lean declarations (relative to the namespace `DirectedReciprocity`) and source file establishing that result. Each new result is followed by an *Outline of the formal proof* explaining the mathematical structure of the machine-checked proof faithfully. As verified automatically by `lean/PaperAxioms.lean` (Appendices A and B), every cited declaration of our library—and every declaration of earlier libraries named in the text—depends only on the three standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`; no file uses `sorry`, `admit`, `native_decide`, `unsafe`, or any custom `axiom`. Finite case checks in Section 3 are discharged inside Lean's kernel via `decide`, `decide +kernel`, or `decide +revert`. The standalone Python script `tools/reproduce_numerics.py` is used solely to cross-check illustrative numerical remarks in Remarks 4.3 and 5.7, never inside any proof.

3 Positional Nesting, Digon Cycles, and Theorem S*

The multi-record nested elimination engine of [10] certifies the polymatroid half-rate bound $2h(V) \le \sum_{v \in V} h(\{v\})$ on a finite recovery relation (V, N) by peeling V into certified pieces (Section 2). In [10], Theorem N3 constructed piece decompositions for symmetric (undirected) triangle-free graphs of maximum degree 3, Theorem U3 extended this to loopless TT-free relations with subcubic underlying graph, and Theorem T constructed them for loopless TT-free relations in which the vertices with at least three parents form an acyclic set; the intermediate target named in item 2 of [10, Section 9.2] asked whether piece decompositions extend to the full directed class $\mathcal{S}$ of Theorem S of [8, Section 6.6] (the all-code version of the fourth paper's Theorem S on its whole class, lying between Theorem U3 and Conjecture P, numbered Conjecture 9.1 there, which remains open), where reciprocal pairs and pure one-way arcs coexist freely subject to reciprocal closure at vertices of in-degree 3, while item 6 of [10, Section 9.2] (see also [10, Remarks 7.3 and 7.5]) asked whether unbounded record counts m can rescue obstructed cores such as X_{15}'s hexagon or supply global elimination certificates on cubic graphs such as the Petersen graph.

In this section we resolve all three questions. We first give a stateless, purely positional reformulation of relational nestedness (Proposition 3.1) that turns impossibility proofs across unbounded record counts $m \ge 1$ into finite-order locking arguments. We then establish exact piece-certificate dichotomies for stride-s hub cores (Proposition 3.2 and Corollary 3.3) and arbitrary chordless digon cycles (Theorem 3.4), classify global record counts across generalized Petersen graphs (Proposition 3.5), and prove **Theorem S*** (Theorem 3.6), **Corollary B*** (Corollary 3.7), and the 21-vertex sharpness witness $\mathrm{G}_{21} \in \mathcal{S}$ (Proposition 3.8).

3.1 Positional characterization of nested elimination

Fix a finite vertex set V, a recovery relation $N : V \to \mathcal{P}(V)$, a context $U \subseteq V$, an index set I of records $P_r \subseteq V$ $(r \in I)$, a supplying-record map $b : V \to I$, an optional outside-parent map $\mathrm{par} : V \to \mathrm{Option}\ V$, an outside set $W \subseteq V$, and a list $l = [v_0, \ldots, v_{k-1}]$ of vertices. For $0 \le j \le k$, write $S_j = \{v_0, \ldots, v_{j-1}\}$ (`HalfRate.rev` from $\emptyset$) for the prefix revealed strictly before step j.

Proposition 3.1 (Positional characterization of nested supply)**.** Lean: `relCharged_of_nested`, `disjoint_nested_iff`, `relCharged_iff_nested`, `PieceCert.mem_supply_of_idxOf_lt`, `PieceCert.digon_ne` [`PositionalNest.lean`]

Let $V, N, U, P, b, \mathrm{par}, W, l$ be as above.

1. (Exact positional equivalence.) *The stateful predicate* `HalfRate.RelCharged` *applied to* $(N, U, P, b, \mathrm{par}, W, \emptyset, \lambda_.\,\emptyset, l)$ *holds if and only if:*
   (a) *for every* $0 \le j < k$ *with* $\mathrm{par}(v_j) = \mathrm{none}$, $N(v_j) \subseteq U \cup S_j \cup P_{b(v_j)}$;
   (b) *for every* $0 \le j < k$ *and* $w \in V$ *with* $\mathrm{par}(v_j) = \mathrm{some}\ w$, $w \in W$ *and* $N(v_j) \subseteq U \cup \{w\} \cup S_j \cup P_{b(v_j)}$; *and*
   (c) *for every* $0 \le j < j' < k$ *and* $w \in V$ *with* $\mathrm{par}(v_j) = \mathrm{par}(v_{j'}) = \mathrm{some}\ w$,
   $$P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}.$$
2. (Disjoint-record simplification.) *Suppose the records* $\{P_r\}_{r \in I}$ *are pairwise disjoint, and let* $0 \le j, j' < k$. *Then* $P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$ *holds if and only if either* $b(v_j) = b(v_{j'})$ *or* $P_{b(v_j)} \subseteq S_{j'}$.
3. (First-appearance supply forcing and digon proper coloring.) *Let c be a piece certificate (*`HalfRate.PieceCert`*) for $H \subseteq V$ in context U with $U \cap H = \emptyset$, and write $\tau(u) = \mathrm{idxOf}(u, c.l)$ for $u \in H$. If $u, v \in H$ satisfy $v \in N(u)$ and $\tau(u) < \tau(v)$, then $v \in c.P_{c.b(u)}$ and $c.b(u) = c.a(v)$. Consequently, for any two distinct vertices $u, v \in H$ forming a digon ($u \in N(v)$ and $v \in N(u)$), their own records are distinct: $c.a(u) \ne c.a(v)$.*

Outline of the formal proof. For (1), induction along l with general initial revealed set S_0 and state map L_0 (`relCharged_step_none`, `relCharged_step_some`, `relCharged_step_nested`) shows that if `HalfRate.RelCharged` holds, then at step j' charged to w after an earlier step $j < j'$ charged to w, the state $L(w)$ contains $\{v_j\} \cup S_j \cup P_{b(v_j)}$ by monotonicity along intermediate updates, forcing $P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$. Conversely (`relCharged_of_nested`), because $v_j \in S_{j'}$ and $S_j \subseteq S_{j'}$ hold automatically whenever $j < j'$, the inclusion $P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$ upgrades to $\{v_j\} \cup S_j \cup P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$, preserving the invariant that $L(w)$ equals either $L_0(w)$ or $\{v_{j_*}\} \cup S_{j_*} \cup P_{b(v_{j_*})}$ for the latest index $j_* < j'$ charged to w. Specializing to $S_0 = \emptyset$ and $L_0 \equiv \emptyset$ gives `relCharged_iff_nested`.

For (2) (`disjoint_nested_iff`), if $b(v_j) \ne b(v_{j'})$, then $P_{b(v_j)} \cap P_{b(v_{j'})} = \emptyset$ by pairwise disjointness, so $P_{b(v_j)} \subseteq S_{j'} \cup P_{b(v_{j'})}$ forces every $x \in P_{b(v_j)}$ into $S_{j'}$; the converse is immediate.

For (3), let $j_u = \tau(u) < \tau(v)$. By definition of first appearance (`not_mem_rev_empty_of_lt_idxOf`), $v \notin S_{j_u}$. Since $v \in H$ and $U \cap H = \emptyset$, we have $v \notin U$ and $v \notin V \setminus (U \cup H)$. Applying (1a) or (1b) at index j_u to $v \in N(u)$ leaves only $v \in c.P_{c.b(u)}$. Because $v \in c.P_{c.a(v)}$ and the records of c are pairwise disjoint, $c.b(u) = c.a(v)$ (`PieceCert.mem_supply_of_idxOf_lt`). If $u \ne v$ form a digon in H, then $\tau(u) \ne \tau(v)$; assuming $\tau(u) < \tau(v)$ by symmetry gives $c.b(u) = c.a(v)$, whereas $c.a(u) \ne c.b(u)$, so $c.a(u) \ne c.a(v)$ (`PieceCert.digon_ne`). □

3.2 Stride-s cores, chordless digon cycles, and generalized Petersen graphs

Let $s \geq 3$ and $k \geq 1$. Define the canonical *stride-s hub relation* `strideRel`(s,k) on $\mathbb{Z}_{sk} \sqcup \mathbb{Z}_s$ by giving each cycle vertex $i \in \mathbb{Z}_{sk}$ parents $\{i-1, i+1\} \subseteq \mathbb{Z}_{sk}$ together with the outside hub $w_{i \bmod s} \in \mathbb{Z}_s$, and giving hubs empty parent sets; let $H_{s,k} = \mathbb{Z}_{sk} \subseteq \mathbb{Z}_{sk} \sqcup \mathbb{Z}_s$ (`strideCore`) be the cycle core. More generally, in any recovery relation (V, N) with context $U \subseteq V$, an injective cycle $f : \mathbb{Z}_L \hookrightarrow V$ with image $H = f(\mathbb{Z}_L)$ is a *stride-3 cycle* in context U (`IsStride3Cycle`) if $3 \mid L$ and there exist three vertices $w_0, w_1, w_2 \in V \setminus (U \cup H)$ such that $w_{i \bmod 3} \in N(f(i))$ for every $i \in \mathbb{Z}_L$.

Proposition 3.2 (Stride-s digon cycle dichotomy)**.** Lean: `stride3_no_pieceCert`, `strideRel`, `strideCore`, `strideCore3_no_pieceCert`, `strideGe4_pieceCert`, `strideEven_pieceCert_two`, `strideCore_pieceCert_iff` [`StrideCycles.lean`]

1. (Unbounded-m stride-3 impossibility.) *Let (V, N) be a finite recovery relation, $U, H \subseteq V$ disjoint subsets, $k \geq 1$, and $f : \mathbb{Z}_{3k} \hookrightarrow V$ a bijection onto H such that $f(i \pm 1) \in N(f(i))$ for all $i \in \mathbb{Z}_{3k}$. If there exist $w_0, w_1, w_2 \in V \setminus (U \cup H)$ with $w_{i \bmod 3} \in N(f(i))$ for all $i \in \mathbb{Z}_{3k}$, then H admits* no *piece certificate (`HalfRate.PieceCert` N U H) over any number of records $m \in \mathbb{N}$.*

2. (Stride-s threshold at $s = 4$.) *For all integers $s \geq 3$ and $k \geq 1$, the canonical stride-s core $H_{s,k}$ admits a piece certificate in context $\emptyset$ under* `strideRel`(s,k) *(for some $m \in \mathbb{N}$) if and only if $s \geq 4$. Moreover, $m = 2$ co-supplied records suffice whenever $s \geq 4$ is even, and $m = 3$ co-supplied records suffice for every $s \geq 4$.*

Outline of the formal proof. For (1) (`stride3_no_pieceCert`), suppose c is an m-record piece certificate for H in context U, and write $\tau(i) = \mathrm{idxOf}(f(i), c.l)$. Because $w_{i \bmod 3} \in N(f(i))$ lies outside $U \cup H$, step $\tau(i)$ must designate $\mathrm{par}(f(i)) =$ some $w_{i \bmod 3}$. We first establish the *Locking Lemma*: if $f(u) \in N(f(i))$, $j \equiv i \pmod 3$, and $\tau(i) \leq \tau(j) < \tau(u)$, then $c.b(f(j)) = c.a(f(u))$. Indeed, Proposition 3.1(3) gives $f(u) \in c.P_{c.b(f(i))}$ and $c.b(f(i)) = c.a(f(u))$; if $\tau(i) < \tau(j)$, then step $\tau(i)$ and step $\tau(j)$ share outside parent $w_{i \bmod 3}$, so Proposition 3.1(1c,2) forces either $c.b(f(i)) = c.b(f(j))$ or $c.P_{c.b(f(i))} \subseteq S_{\tau(j)}$, and the latter is impossible because $f(u) \in c.P_{c.b(f(i))}$ has $\tau(j) < \tau(u)$ and so is unrevealed at $\tau(j)$. As an immediate consequence, if $j_1 \equiv j_2 \pmod 3$ satisfy $f(u_1) \in N(f(j_1))$, $f(u_2) \in N(f(j_2))$, and $\max(\tau(j_1), \tau(j_2)) < \min(\tau(u_1), \tau(u_2))$, applying locking at $\max(\tau(j_1), \tau(j_2))$ yields $c.a(f(u_1)) = c.a(f(u_2))$.

Now let $z \in \mathbb{Z}_{3k}$ maximize $\tau(z)$ over all of $\mathbb{Z}_{3k}$, set $r_0 = z \bmod 3$, and let x maximize $\tau(x)$ among $\{i \in \mathbb{Z}_{3k} : i \not\equiv r_0 \pmod 3\}$. Let $r_x = x \bmod 3 \neq r_0$ and let $r_3 \in \{0,1,2\} \setminus \{r_0, r_x\}$ be the remaining residue class modulo 3. Since $f(y \pm 1)$ realize the two residues distinct from y mod 3 for every $y \in \mathbb{Z}_{3k}$, x has a cycle neighbor $u_3 \equiv r_3 \pmod 3$ and z has cycle neighbors $v_3 \equiv r_3 \pmod 3$ and $v_x \equiv r_x \pmod 3$. Because $u_3, v_3 \not\equiv r_0 \pmod 3$ and neither lies in r_x, the choice of x gives $\tau(u_3) < \tau(x) < \tau(z)$ and $\tau(v_3) < \tau(x) < \tau(z)$. Since $u_3 \equiv v_3 \equiv r_3 \pmod 3$, the two-arc corollary forces $c.a(f(x)) = c.a(f(z))$. If $x = v_x$, then $f(x)$ and $f(z)$ form a digon along the cycle, contradicting $c.a(f(x)) \neq c.a(f(z))$ (Proposition 3.1(3)). Otherwise $v_x \neq x$ lies in $r_x \neq r_0$, so $\tau(v_x) < \tau(x) < \tau(z)$; locking along $f(z) \in N(f(v_x))$ from v_x to x forces $c.b(f(x)) = c.a(f(z)) = c.a(f(x))$, contradicting $c.a(f(x)) \neq c.b(f(x))$.

For (2), when $s = 3$, (1) rules out any `HalfRate.PieceCert` (`strideCore3_no_pieceCert`). When $s \geq 4$ is even (`strideEven_pieceCert_two`), $2 \mid s \mid sk$, so partitioning $H_{s,k}$ by parity into two records P_0, P_1 and eliminating all even indices before all odd indices makes $b(i) = 1 - (i \bmod 2)$ constant on each hub fiber $i \bmod s$ (co-supplied). For general $s \geq 4$ (`strideGe4_pieceCert`), pulling back the proper 3-coloring `HalfRate.ccol`(s) of $\mathbb{Z}_s$ along $i \mapsto i \bmod s$ partitions $H_{s,k}$ into three records P_0, P_1, P_2; on each residue class $i \equiv r \pmod s$, both neighbors $i \pm 1$ have colors

$(r \pm 1) \bmod s$ in $\mathbb{Z}_s$, and as in Theorem N3 of [10, Theorem 4.3] (`HalfRate.cyclePiece`; see also [10, Lemma 5.7(b)]), ordering the fibers so that one neighbor color is already revealed while the other serves as the constant supplying record $b(i)$ on fiber r yields an explicit 3-record co-supplied certificate. □

Applying Proposition 3.2(1) to the 15-vertex relation X_{15} of [10, Section 7.2, Remark 7.3] (`HalfRate.x15`), whose unique minimum nonempty core is the induced 6-cycle $\{0,1,2,3,4,5\}$ (see [10, Remark 7.3], Lean-verified as `HalfRate.x15_cores`) attached to outside vertices $6,7,8$ with stride 3, immediately settles the unbounded-m core question of [10, Remark 7.3; Section 9.2, item 6]:

Corollary 3.3 (Unbounded-m obstruction on X_{15})**.** Lean: `x15_hexagon_no_pieceCert` [`ClassSExamples.lean`]

The unique minimum nonempty core $\{0,1,2,3,4,5\}$ *of* X_{15} *admits no piece certificate* `HalfRate.PieceCert HalfRate.x15` $\emptyset$ $\{0,1,2,3,4,5\}$ *for any number of records* $m \geq 1$.

Outline of the formal proof. Instantiate `stride3_no_pieceCert` with $k = 2$, $f = [0,1,2,3,4,5]$, and outside hub triple $w = [6,7,8]$; all finite membership and injectivity side conditions discharge by `decide`. □

We next show that on *any* chordless digon cycle C_L ($L \geq 4$) in a TT-free relation where each cycle vertex has at most one active outside parent, the stride-3 hub configuration is the *unique* obstruction to piece certifiability across all $m \geq 1$.

Theorem 3.4 (Universal chordless digon cycle dichotomy)**.** Lean: `IsStride3Cycle`, `not_pieceCert_of_isStride3Cycle`, `exists_non_stride3_pos`, `digonCycle4_pieceCert`, `digonCycleGe5_pieceCert`, `digonCycle_pieceCert_of_not_stride3`, `digonCycle_pieceCert_iff` [`DigonCycle.lean`]

Let (V, N) *be a finite recovery relation,* $U \subseteq V$ *a context,* $L \geq 4$*, and* $f : \mathbb{Z}_L \hookrightarrow V$ *an injective map with image* $H = f(\mathbb{Z}_L)$ *disjoint from* U*. Assume that:*

1. *no outside vertex* $w \in V \setminus (U \cup H)$ *is a common parent of two consecutive cycle vertices* $f(i)$ *and* $f(i+1)$ *(local TT-freeness);*
2. $f(i \pm 1) \in N(f(i))$ *for every* $i \in \mathbb{Z}_L$*, and conversely* $f(j) \in N(f(i)) \implies j \in \{i-1, i+1\}$ *(chordless digon cycle); and*
3. *each* $f(i)$ *has at most one active parent in* $V \setminus (U \cup H)$*.*

Then H *admits a piece certificate (*`HalfRate.PieceCert` N U H*) if and only if* f *is not a stride-3 cycle in context* U *(*`IsStride3Cycle` N U f *fails). When* `IsStride3Cycle` N U f *fails, there exists an explicit piece certificate with* $m = 2$ *records for* $L = 4$ *and* $m \leq L + 1$ *records for* $L \geq 5$*.*

Outline of the formal proof. Necessity (`not_pieceCert_of_isStride3Cycle`) follows directly from Proposition 3.2(1). For sufficiency (`digonCycle_pieceCert_of_not_stride3`), write $\mathrm{op}(i) \in \mathrm{Option}\ V$ for the unique outside parent of $f(i)$ in $V \setminus (U \cup H)$ (or none if none exists).

When $L = 4$ (`digonCycle4_pieceCert`), partition H into $P_0 = \{f(0), f(2)\}$ and $P_1 = \{f(1), f(3)\}$, set $b = 1 - a$, and eliminate in order $[f(0), f(2), f(1), f(3)]$. By (1), $\mathrm{op}(i)$ cannot coincide on adjacent cycle vertices, so any shared outside parent occurs on opposite vertices $\{0,2\}$ or $\{1,3\}$, where b is constant.

When $L \geq 5$ (`digonCycleGe5_pieceCert`), we first prove (`exists_non_stride3_pos`) that failure of `IsStride3Cycle` N U f yields an index $s_0 \in \mathbb{Z}_L$ such that $\mathrm{op}(s_0) = \text{none}$ or $\mathrm{op}(s_0) \neq$

$\mathrm{op}(s_0+3)$: otherwise every $i \in \mathbb{Z}_L$ has $\mathrm{op}(i) =$ some w_i periodic of period 3 along $\mathbb{Z}_L$, forcing $3 \mid L$ (since $3 \nmid L$ would make op constant on $\mathbb{Z}_L$, violating (1)) and exhibiting w_0, w_1, w_2 witnessing `IsStride3Cycle`. Rotating f so that $s_0 = 2$ gives $\mathrm{op}(2) =$ none or $\mathrm{op}(2) \neq \mathrm{op}(5 \bmod L)$. We then eliminate H over $L+1$ records $\{0, 1, \ldots, L\}$ in the two-sided order

$$[f(1),\ f(2),\ f(0),\ f(L-1),\ f(L-2),\ \ldots,\ f(4),\ f(3)]$$

(`dcPerm`), assigning own records (`dcCol`) and supplying records (`dcSupply`) as follows:

- $f(0)$ and $f(2)$ have own record 0; $f(1)$ has own record 1 and supplying record 0 (covering both neighbors $f(0), f(2)$ at step 0);
- $f(3)$ has own record 3, and for $k \in \{4, \ldots, L-1\}$ we set $a(f(k)) = 3$ if $\mathrm{op}(2) =$ some $w = \mathrm{op}(k+1 \bmod L)$ and $a(f(k)) = k$ otherwise. By our rotation $\mathrm{op}(5 \bmod L) \neq \mathrm{op}(2)$ (when $\mathrm{op}(2)$ is some w), we always have $a(f(4)) = 4 \neq 3$, and by (1) no two consecutive $k, k+1$ can both merge into record 3;
- at step 1, $f(2)$ uses supplying record 3 (covering $f(3)$, with $f(1)$ already revealed); for steps $j \in \{2, \ldots, L-2\}$ eliminating $f(k)$ ($k \in \{0, L-1, \ldots, 4\}$), its successor $f(k+1 \bmod L)$ is already revealed and we set $b(f(k)) = a(f(k-1 \bmod L))$ covering its predecessor; at step $L-1$, $f(3)$ has both neighbors $f(2), f(4)$ already revealed and uses the empty auxiliary record L.

To verify nestedness (Proposition 3.1(1c)), suppose $0 \le j < j' < L$ share op $=$ some w. Step 0 ($f(1)$) has $P_{b(f(1))} = P_0 = \{f(0), f(2)\}$; by (1) neither $f(2)$ ($j' = 1$) nor $f(0)$ ($j' = 2$) can share $\mathrm{op}(1)$, and for all $j' \ge 3$ we already have $P_0 \subseteq S_3 \subseteq S_{j'}$. Step 1 ($f(2)$) has $b(f(2)) = 3$; by (1) the final vertex $f(3)$ ($j' = L-1$) cannot share $\mathrm{op}(2)$, while for any $2 \le j' \le L-2$ with $\mathrm{op}(k) = \mathrm{op}(2) =$ some w, our definition of `dcCol` placed $f(k-1 \bmod L)$ into record 3, so $b(f(k)) = 3 = b(f(2))$ (co-supplied). Finally, for $2 \le j < j' < L$, either $b(v_j) = 3$, in which case $\mathrm{op}(v_j) = \mathrm{op}(2) =$ some $w = \mathrm{op}(v_{j'})$, excluding $j' = L-1$ by (1) and forcing $b(v_{j'}) = 3$ for $j' \le L-2$, or else $b(v_j) = k-1 \ge 4$ is a singleton $\{f(k-1)\}$ revealed immediately at step $j+1$, whereas $j' = j+1$ cannot share w with j by (1), giving $P_{b(v_j)} \subseteq S_{j+2} \subseteq S_{j'}$. □

Before turning to $\mathcal{S}$, we resolve the global-certificate question of [10, Remark 7.5; Section 9.2, item 6] by determining exact global record thresholds across the cubic generalized Petersen family $\mathrm{GP}(m_{\mathrm{gp}}, 2)$ ($m_{\mathrm{gp}} \ge 5$, `gpGraph`), on vertex set $\{0, \ldots, 2m_{\mathrm{gp}} - 1\}$ with outer rim edges $i \leftrightarrow (i \pm 1 \bmod m_{\mathrm{gp}})$, inner star edges $m_{\mathrm{gp}} + i \leftrightarrow m_{\mathrm{gp}} + (i \pm 2 \bmod m_{\mathrm{gp}})$, and spokes $i \leftrightarrow m_{\mathrm{gp}} + i$.

Proposition 3.5 (Non-monotone global record hierarchy on $\mathrm{GP}(m_{\mathrm{gp}}, 2)$)**.** Lean: `nonempty_elimCert_iff_pieceCert_univ`, `gpGraph`, `gp52_eq_petersenGraph`, `gp52_no_elimCert`, `petersen_no_pieceCert_univ`, `gp72_no_elimCert_le3`, `gp72_elimCert4`, `gp92_elimCert3`, `generalizedPetersen_hierarchy` [`PetersenHierarchy.lean`]

1. (Global equivalence.) *For any finite recovery relation (V, N) and $m \in \mathbb{N}$, there exists a global piece certificate p of type* `HalfRate.PieceCert` $N\ \emptyset\ V$ *with $p.m = m$ if and only if* `HalfRate.ElimCert` $N\ m$ *is nonempty.*
2. ($\mathrm{GP}(5,2)$, Petersen graph, $n = 10$.) $\mathrm{GP}(5,2)$ *coincides with* `MotionDepthCodes.petersenGraph` *and admits* no *global elimination certificate (of type* `HalfRate.ElimCert` (nbhd $\mathrm{GP}(5,2)$) m*, or equivalently global* `HalfRate.PieceCert`*) for* any *number of records $m \in \mathbb{N}$.*

3. ($\mathrm{GP}(7,2)$, $n = 14$.) $\mathrm{GP}(7,2)$ *admits no global elimination certificate with* $m \le 3$ *records, strictly separating* $m = 3$ *from* $m = 4$*, and admits an explicit* 4*-record global elimination certificate.*

4. ($\mathrm{GP}(9,2)$, $n = 18$.) $\mathrm{GP}(9,2)$ *drops back from* 4 *records to an explicit* 3*-record global elimination certificate.*

Outline of the formal proof. For (1) (`nonempty_elimCert_iff_pieceCert_univ`), when $U = \emptyset$ and $H = V$, the outside set $W = V \setminus V = \emptyset$ forces every step of a piece certificate p to have $\mathrm{par}(v_j) = \mathrm{none}$; given any proper subset $S \subsetneq V$, taking the first index j of $p.l$ with $v_j \notin S$ gives $S_j \subseteq S$ and hence $N(v_j) \subseteq S \cup P_{b(v_j)}$, yielding `HalfRate.ElimCert` N $p.m$. Conversely, greedy extraction from `HalfRate.ElimCert` N m via `HalfRate.exists_relCharged` produces a global `HalfRate.PieceCert` with m records.

By Proposition 3.1(3) (`elimCert_adj_ne`), any m-record global elimination certificate on an undirected graph G induces a proper coloring $a : V \to \{0, \dots, m-1\}$ into independent sets; moreover, in an elimination certificate, no proper subset $S \subsetneq V$ can be *stuck*, meaning that every $v \notin S$ has two neighbors outside S of distinct a-colors (since a single supplying record $P_{b(v)}$ can belong to only one color class). For (2) (`gp52_no_elimCert`), replacing $a(u)$ by the minimum vertex index in its own record maps any arbitrary-m global certificate on $\mathrm{GP}(5,2)$ to a canonical idempotent proper partition of $\{0, \dots, 9\}$, and an exhaustive kernel-checked Boolean certificate over all canonical proper partitions exhibits a stuck proper subset $S \subsetneq \{0, \dots, 9\}$ in every case (`decide +kernel`). For (3) (`gp72_no_elimCert_le3`, `gp72_elimCert4`), any $m \le 3$ certificate embeds into a normalized proper 3-coloring of $\mathrm{GP}(7,2)$, every one of which admits a kernel-verified stuck subset, while an explicit 4-coloring and 14-step schedule verify `HalfRate.ElimCert` (nbhd $\mathrm{GP}(7,2)$) 4 by `decide`. For (4) (`gp92_elimCert3`), an explicit 3-coloring and 18-step schedule verify `HalfRate.ElimCert` (nbhd $\mathrm{GP}(9,2)$) 3 by `decide`. □

3.3 Piece decomposition and barrier bound for $\mathcal{S}$

Recall from [8, Section 6.6] that a finite recovery relation (V, N) belongs to $\mathcal{S}$ (`IsClassS`) if N is irreflexive, TT-free (`HalfRate.TTFree`), has maximum in-degree at most 3 ($|N(v)| \le 3$ for all $v \in V$), and satisfies *reciprocal closure at full in-degree*:

$$\forall v \in V, \quad |N(v)| = 3 \implies \forall u \in V,\ v \in N(u) \implies u \in N(v).$$

Equivalently, vertices of in-degree 3 may have only reciprocal incident arcs (hence out-degree 3), whereas vertices of in-degree at most 2 may send arbitrarily many pure one-way outgoing arcs. More generally, (V, N) satisfies relative $\mathcal{S}$ on $R \subseteq V$ (`ClassSOn`) if the internal restriction $N|_R$ (`resRel`) belongs to $\mathcal{S}$ and N is TT-free on R. Note that `ClassSOn` is hereditary under passing to arbitrary subsets $R' \subseteq R$.

Theorem 3.6 (Piece decomposition and half-rate bound for $\mathcal{S}$)**.** Lean: `IsClassS`, `ClassSOn`, `exists_inClosed_scc`, `liftPieceCertInClosed`, `exists_piece_of_scc_classS`, `classSOn_decomp`, `classS_decomp`, `rateBound_classS`, `card_sq_le_classS` [`ClassS.lean`]

Let (V, N) *be a finite recovery relation.*

1. *For every subset* $R \subseteq V$ *satisfying* `ClassSOn` N R*,* (V, N) *admits a certified piece decomposition (*`HalfRate.Decomp` N $(V \setminus R)$ pcs*) from context* $V \setminus R$*.*

2. *In particular, if* $N \in \mathcal{S}$*, then* (V, N) *admits a certified piece decomposition from* $\emptyset$*, satisfies* `MotionDepthCodes.RateBound` N $(1/2)$ *(*$2h(V) \le \sum_{v \in V} h(\{v\})$ *for every valid polymatroid* h*),*

and every storage code $C \subseteq \alpha^V$ (linear or nonlinear, over any nonempty finite alphabet α) satisfies

$$|C|^2 \le |\alpha|^{|V|}.$$

Outline of the formal proof. By strong induction on $|R|$ (`classSOn_decomp`), if R is nonempty, condensation of the reachability relation on R yields a nonempty *in-closed* strongly connected component $S \subseteq R$ (`exists_inClosed_scc`): every parent in R of a vertex $v \in S$ already lies in S, and every $u, v \in S$ reach one another via directed paths in $M := N|_S$. Because S is in-closed inside R, any piece certificate for $H \subseteq S$ in empty context under M lifts immediately to a piece certificate for H in context $V \setminus R$ under N (`liftPieceCertInClosed`), after which induction applies to $R \setminus H$.

It therefore suffices to find a nonempty certified piece $H \subseteq S$ in context $\emptyset$ for a strongly connected relation $M \in \mathcal{S}$ supported on S (`exists_piece_of_scc_classS`). Let $\mathcal{D} = \texttt{digonGraph}(M)$ be the undirected simple graph of reciprocal pairs on S; since M is TT-free, $\mathcal{D}$ is triangle-free (`digonGraph_cliqueFree3`). Write $W_S = \{v \in S : |M(v)| = 2\}$.

1. *Low in-degree vertex.* If some $v \in S$ has $|M(v)| \le 1$, then $H = \{v\}$ is certified by two records $\{v\}$ and $\emptyset$ (`HalfRate.singlePiece`), uncharged if $|M(v)| = 0$ or charged to its sole parent if $|M(v)| = 1$.

2. *Directed cycle in W_S.* Assume henceforth $|M(v)| \in \{2, 3\}$ on S. If W_S is not M-acyclic, let C be a shortest directed cycle of M inside W_S. Every vertex of C has in-degree 2 in M, hence one cycle parent and at most one outside parent, and minimality plus TT-freeness make C chordless with no shared outside parent on consecutive vertices; if some i has $f(i-1) \in M(f(i))$ and $f(i) \in M(f(i-1))$, minimality forces $|C| = 2$ (a digon of in-degree-2 vertices), while otherwise C is a pure one-way chordless directed cycle of length ≥ 3, both certified by $m = 2$ records (`HalfRate.i5Piece`, as in Theorem T of [10]).

3. *Acyclic digon graph.* Assume W_S is M-acyclic and $\mathcal{D}[S]$ is a forest. Fix a proper 2-coloring $\kappa : S \to \{0, 1\}$ of $\mathcal{D}[S]$. Neither color class $S_k = \{v \in S : \kappa(v) = k\}$ can contain a directed cycle of M: any directed cycle inside S_k cannot lie wholly in W_S (since W_S is M-acyclic), so it must visit some $v \in S_k$ with $|M(v)| = 3$, whose predecessor $u \in S_k$ on the cycle satisfies $v \in M(u)$ and hence $u \in M(v)$ by $\mathcal{S}$, forming a monochromatic digon inside S_k. Thus each S_k is M-acyclic, so printing S_0 and S_1 in topological order with $b = 1 - \kappa$ gives a global 2-record certificate on $H = S$.

4. *Shortest digon cycle and hub augmentation.* Finally, suppose $\mathcal{D}[S]$ contains cycles, and let $g \ge 4$ be the girth of $\mathcal{D}[S]$.

   - *Subcase 4a (minimal chordal arc).* Suppose some g-cycle c' of $\mathcal{D}[S]$ carries a non-cycle M-arc between vertices of cycle distance d; reversing c' via `cycRev` if necessary, we may assume $c'(i_0) \in M(c'(i_0 + d))$ with $2 \le d \le \lfloor g/2 \rfloor$, and choose (d, c', i_0) minimizing d ($d = 2$ is ruled out by `HalfRate.TTFree`, so $3 \le d$). Consider the $(d+1)$-vertex arc $H = \{p_0, \ldots, p_d\}$ where $p_s = c'(i_0 + s)$. Minimality of g makes $p_0 \in M(p_d)$ a *pure one-way* arc ($p_d \notin M(p_0)$), so $\mathcal{S}$ forces $|M(p_0)| \le 2$ (p_1 plus at most one outside parent), while every p_s ($1 \le s \le d$) already has two internal parents in H (p_{s-1} and p_{s+1} for $s < d$, or p_{d-1} and p_0 for $s = d$) and hence at most one outside parent. Minimality of d rules out all other internal M-arcs on H, and eliminating $[p_0, p_1, \ldots, p_d]$ in forward order with singleton records $P_s = \{p_s\}$ ($0 \le s \le d$), empty record $P_{d+1} = \emptyset$, and $b(p_s) = s + 1$ yields an explicit $(d+2)$-record nested certificate on H (`minChordArc_pieceCert`).

- *Subcase 4b (chordless non-stride-3 cycle).* Suppose no g-cycle of $\mathcal{D}[S]$ has any non-cycle M-chord. Pick any g-cycle $C = \{c(0), \dots, c(g-1)\}$. If C is not a stride-3 cycle in context $\emptyset$ under M, Theorem 3.4 immediately certifies $H = C$.
- *Subcase 4c (hub augmentation of a stride-3 cycle).* Suppose C is a stride-3 cycle ($3 \mid g$, hence $g \geq 6$ by triangle-freeness) with hubs $w_0, w_1, w_2 \in S \setminus C$. Because S is strongly connected and $w_0 \in S \setminus C$, some vertex $c(i_0) \in C$ sends an outgoing M-arc to some $y \in S \setminus C$, i.e., $c(i_0) \in M(y)$. Since every vertex of C has $|M(c(i))| = 3$ (two cycle neighbors plus $w_{i \bmod 3}$), reciprocal closure in $\mathcal{S}$ forces $y \in M(c(i_0))$, so $y = w_{r_0}$ where $r_0 = i_0 \bmod 3$. Now w_{r_0} forms a digon with $c(i_0)$ and cannot form a digon with any other $c(k)$ ($k \neq i_0$), as two digons from w_{r_0} to C would create a $\mathcal{D}$-cycle of length at most $\lfloor g/2 \rfloor + 2 < g$ (or a 4-cycle when $g = 6$). Because $w_{r_0} \in M(c(i_0 + 3))$ and $i_0 + 3 \not\equiv i_0 \pmod{g}$, $c(i_0 + 3) \notin M(w_{r_0})$, so $\mathcal{S}$ forces $|M(w_{r_0})| = 2$: w_{r_0} has parent $c(i_0)$ on C and at most one parent outside $H := \{w_{r_0}\} \cup C$ (which may even be another hub w_r). Rotating C so $i_0 = 0$ and $r_0 = 0$, the $(g+1)$-vertex augmented piece $H = \{w_0\} \cup C$ admits an explicit $(g+2)$-record sequential nested certificate (`augmentedStride3_pieceCert`): list H in cyclic order $[v_0, \dots, v_g] = [w_0, c(0), c(1), \dots, c(g-1)]$, assign own-record indices $(g, 1, 0, 3, 4, \dots, g-1, 0)$ with empty auxiliary record $P_{g+1} = \emptyset$, and set sequential supply $b(v_j) = j + 1$ for $0 \leq j \leq g$, verifying nestedness across shared outside parents via Proposition 3.1(1).

Applying `HalfRate.rateBound_of_decomp` and `MotionDepthCodes.card_sq_le_of_rateBound_half` completes (2). □

For any vertex cut $S \subseteq V$ of (V, N), write $I_N(S) = \{v \in V \setminus S : N(v) \subseteq S\}$ (`isolatedOutDir`) for the set of vertices outside S all of whose active recovery parents lie inside S. Applying Theorem 3.6(1) in context $U = S \cup I_N(S)$ yields the directed barrier bound for $\mathcal{S}$:

Corollary 3.7 (Directed barrier inequality for $\mathcal{S}$)**.** Lean: `isolatedOutDir`, `barrier_ineq_classS`, `log_card_le_barrier_classS` [`ClassS.lean`]

Let (V, N) be a finite recovery relation in $\mathcal{S}$, and let $S \subseteq V$.

1. *Every polymatroid h valid for N satisfies*

$$2h(V) \leq 2 \sum_{v \in S} h(\{v\}) + \sum_{v \in V \setminus (S \cup I_N(S))} h(\{v\}).$$

2. *Every storage code $C \subseteq \alpha^V$ over a nonempty finite alphabet α satisfies*

$$2 \log|C| \leq (|V| + |S| - |I_N(S)|) \log|\alpha|.$$

Outline of the formal proof. Let $U = S \cup I_N(S)$. By Theorem 3.6(1), $V \setminus U$ admits a piece decomposition (`HalfRate.Decomp` N U pcs), so `HalfRate.decomp_ineq` gives $2(h(V) - h(U)) \leq \sum_{v \notin U}(h(U \cup \{v\}) - h(U)) \leq \sum_{v \notin U} h(\{v\})$. Since every $v \in I_N(S)$ has $N(v) \subseteq S$, validity gives $h(U) = h(S) \leq \sum_{v \in S} h(\{v\})$, proving (1) (`barrier_ineq_classS`); evaluating on the uniform code-entropy polymatroid of C proves (2) (`log_card_le_barrier_classS`). □

Finally, we exhibit a strongly connected, completely triangle-free digraph in $\mathcal{S}$ on which the low-in-degree, W_S-cycle, and shortest-digon-cycle rules all stall simultaneously, whereas hub augmentation (Subcase 4c of Theorem 3.6) decomposes it.

Proposition 3.8 (Simultaneous stalling of earlier rules on $G_{21} \in \mathcal{S}$)**.** Lean: `g21Rel`, `g21_isClassS`, `g21_no_any_triangle`, `g21_min_indeg_and_acyclic_W`, `g21_stronglyConnected`, `g21_hex_no_pieceCert`, `g21_shortest_digon_cycles`, `rateBound_g21` [`ClassSExamples.lean`]

Let G_{21} *(*`g21Rel`*) be the recovery relation on* $\{0,\dots,20\}$ *formed by three disjoint digon* 6*-cycles* $A = \{0,\dots,5\}$, $B = \{6,\dots,11\}$, $C = \{12,\dots,17\}$ *and three hubs* $18, 19, 20$*, where each* $i \in \{0,\dots,17\}$ *has cycle parents* $i \pm 1$ *(modulo* 6 *within its hexagon) plus hub parent* $18 + (i \bmod 3)$*, and the three hubs have parent sets* $N(18) = \{0, 15\}$, $N(19) = \{1, 7\}$, $N(20) = \{8, 14\}$*. Then:*

1. $G_{21} \in \mathcal{S}$*,* G_{21} *is strongly connected, and the underlying undirected support of* G_{21} *is triangle-free;*
2. *every vertex of* G_{21} *has in-degree* 2 *or* 3*, and the in-degree-*2 *set* $W = \{18, 19, 20\}$ *has no internal arcs (so low-in-degree and* W*-cycle rules do not apply);*
3. *every cycle of length at most* 6 *in* `digonGraph`(G_{21}) *has length* 6 *and vertex set* A*,* B*, or* C *(each forming a mutual* 6*-cycle in* G_{21}*), and none of* A*,* B*,* C *admits a piece certificate (*`HalfRate.PieceCert` G_{21} $\emptyset$ H*) for any* $m \in \mathbb{N}$*;*
4. *nevertheless,* G_{21} *admits a piece decomposition and satisfies* `MotionDepthCodes.RateBound` G_{21} $(1/2)$*.*

Outline of the formal proof. Membership $G_{21} \in \mathcal{S}$ (`g21_isClassS`), complete triangle-freeness (`g21_no_any_triangle`), minimum in-degree 2 and independence of $W = \{18, 19, 20\}$ (`g21_min_indeg_and_acyclic_W`), and strong connectivity via an explicit 32-step closed walk visiting all 21 vertices (`g21_stronglyConnected`) are verified in the kernel. Unbounded-m impossibility on A, B, and C (`g21_hex_no_pieceCert`) follows by instantiating Proposition 3.2(1) with hub triple $[18, 19, 20]$. To classify all digon cycles of length $3 \le g \le 6$ (`g21_shortest_digon_cycles`), a kernel-checked 4-step BFS avoiding w verifies that for every hub $w \in \{18, 19, 20\}$, its two digon neighbors lie in distinct hexagons and have no $\mathcal{D}$-walk of length ≤ 4 avoiding w, so no cycle of length $g \le 6$ can visit any hub; projecting any remaining cycle into $\mathbb{Z}_6$ modulo 6 rules out $g \in \{3, 4, 5\}$ by parity and injectivity, leaving A, B, and C. Finally, Theorem 3.6 (peeling, for instance, a 7-vertex hub-augmented piece such as $\{18\} \cup A$ via Subcase 4c) yields `MotionDepthCodes.RateBound` G_{21} $(1/2)$ (`rateBound_g21`). □

4 Scalar Linear Codes at Degree 4 Without Heavy Separation

On 2-head tape computations whose dependency DAG is cut to bounded depth, vertices with up to four active parents arise naturally whenever two heads revisit cells across coupled scales [7, Section 8]. Over an arbitrary field F, let $M \in F^{V \times V}$ have unit diagonal ($M_{v,v} = 1$) and no recovery triangle: there are no three distinct vertices $a, b, c \in V$ with $b \in N_M(a)$, $c \in N_M(b)$ and $c \in N_M(a)$, where $N_M(v) = \{u \ne v : M_{v,u} \ne 0\}$ (`MotionDepthCodes.ThmSHyp`). Recall from Section 2 that $v \in V$ is *light* if it satisfies $(\star\star)$ (`MotionDepthCodes.Star`: either $|N_M(v)| \le 2$, or $|N_M(v)| = 3$ and $v \in N_M(u) \implies u \in N_M(v)$) and *heavy* otherwise, with heavy set $H_M = \{v \in V : \neg(\star\star)(v)\}$ (`BoundedNesting.heavyVerts`), and that $(\star 4)$ (`BoundedNesting.Star4`) requires $|N_M(v)| \le 3$, or $|N_M(v)| = 4$ with reciprocal outgoing recovery arcs.

In [7, Section 8.2, Theorem 8.3], the half-rate bound $\dim_F \ker M \le \lfloor |V|/2 \rfloor$ was proved under $(\star 4)$ assuming either $|H_M| \le 2$ or *both* (S1) (`BoundedNesting.SepS1`: no $v \in V$ has two distinct heavy recovery neighbors $p, q \in N_M(v) \cap H_M$) and (S2) (`BoundedNesting.SepS2`: no $h \in H_M$ has any heavy recovery neighbor in $N_M(h) \cap H_M$). On an undirected triangle-free support graph

$G = (V, E)$ of maximum degree 4, combining (S1) and (S2) forced heavy vertices to lie at graph distance at least 3 pairwise (`BoundedNesting.Far3`), ruling out adjacent heavy pairs. Notice, however, that on a triangle-free graph G, two adjacent vertices $u, v \in H_M$ automatically share no common neighbor in G. It is therefore natural to ask—as discussed in [7, Section 8.2, Remark 8.8; Section 10, Problem 10.4]—whether (S2) is merely an artifact of the proof.

We prove that (S2) can be dropped altogether, both in the directed matrix setting and in the undirected graph setting, and deduce a strictly stronger matching-deficiency bound.

4.1 Theorem SH*: eliminating (S2)

For a subset $W \subseteq V$, write $\mathrm{rs}_W(v) = N_M(v) \cap W$ (`BoundedNesting.rs`), let $(\star\star)_W$, $(\star 4)_W$, and $H_{M,W}$ (`BoundedNesting.StarOn`, `BoundedNesting.Star4On`, `BoundedNesting.heavySet`) be the restrictions of $(\star\star)$, $(\star 4)$, and H_M to W, and write $\mathrm{S1}_W$ (`BoundedNesting.S1On`) for the condition that no $v \in W$ has two distinct recovery neighbors in $\mathrm{rs}_W(v) \cap H_{M,W}$. Define `SHStarOn`$(M, W)$ to hold when every $v \in W$ satisfies $(\star 4)_W$ and either $|H_{M,W}| \le 2$ or $\mathrm{S1}_W$ holds. For an undirected graph $G = (V, E)$, write `NoCommonNbhd`(G, u, v) if there is no $w \in V$ adjacent in G to both u and v.

Theorem 4.1 (Half-rate kernel bound under (S1) alone)**.** Lean: `SHStarOn`, `F3_star`, `odd_state_star`, `MinCE.false_SHStar`, `finrank_subCode_le_half_SHStar`, `theoremSHStar`, `theoremSHStar_transpose`, `NoCommonNbhd`, `theoremSHStar_graph` [`TheoremSHStar.lean`]

Let F be a field, V a finite vertex set, and $M \in F^{V \times V}$.

1. (Hereditary subcode form.) *Suppose M satisfies* `MotionDepthCodes.ThmSHyp` *(unit diagonal and no recovery triangle). For every $W \subseteq V$ satisfying* `SHStarOn`(M, W)*, the restricted code* $\mathrm{subCode}_M(W) = \{x \in F^V : \mathrm{supp}(x) \subseteq W \text{ and } (Mx)|_W = 0\}$ *satisfies*

$$\dim_F \mathrm{subCode}_M(W) \le \lfloor |W|/2 \rfloor.$$

2. (Global matrix and transpose forms.) *Suppose M satisfies* `MotionDepthCodes.ThmSHyp`*. If every $v \in V$ satisfies $(\star 4)$ and either $|H_M| \le 2$ or* `BoundedNesting.SepS1`(M) *holds, then*

$$\dim_F \ker M \le \lfloor |V|/2 \rfloor;$$

the same bound holds if M^{T} satisfies $(\star 4)$ everywhere and either $|H_{M^{\mathsf{T}}}| \le 2$ or `BoundedNesting.SepS1`(M^{T}) *holds.*

3. (Graph form.) *Let $G = (V, E)$ be a triangle-free simple graph of maximum degree at most 4, and suppose $M_{v,v} = 1$ for all $v \in V$ and $M_{v,u} = 0$ whenever $v \ne u$ and $\{v, u\} \notin E$. If either $|H_M| \le 2$ or every pair of distinct $u, v \in H_M$ satisfies* `NoCommonNbhd`(G, u, v)*, then*

$$\mathrm{rank}_F M \ge \lceil |V|/2 \rceil \qquad \text{and} \qquad \dim_F \ker M \le \lfloor |V|/2 \rfloor.$$

Outline of the formal proof. First observe that `SHStarOn`(M, W) is hereditary under passing to subsets $W' \subseteq W$ (`SHStarOn.mono`), since $(\star 4)_{W'}$ and membership in $H_{M,W'}$ are monotone in W. Thus (1) (`finrank_subCode_le_half_SHStar`) reduces by strong induction on $|W|$ to ruling out a minimal counterexample (`BoundedNesting.MinCE` M W) on which `SHStarOn`(M, W) holds (`MinCE.false_SHStar`). Recall from [7, Section 8.2] that a minimal counterexample W has odd cardinality $|W| = 2k + 1$, $\mathrm{cdim}_W(\emptyset) = k + 1$, every proper subset $W' \subsetneq W$ satisfies $\dim_F \mathrm{subCode}_M(W') \le \lfloor |W'|/2 \rfloor$, and every $v \in W$ has $|\mathrm{rs}_W(v)| \ge 2$. For a state $X \subseteq W$ from start vertex $v_0 \in X$, write $\mathrm{pend}_W(X, u) = \mathrm{rs}_W(u) \setminus X$ and $\mathrm{cdim}_W(X) = \dim_F(\mathrm{subCode}_M(W) \cap \mathrm{zeroOn}(X))$.

Where did [7] use (S2)? Precisely in two places: Phase E (when an odd state Z expands along a heavy vertex $h \in Z$ with $|\mathrm{pend}_W(Z,h)| = 3$) and Case 3 of the starting split (when choosing a light neighbor v_0 of a heavy vertex h). Notice that Phases F1 and F2 of [7] (`BoundedNesting.F1`, `BoundedNesting.F2`) already operate under `BoundedNesting.F2Mode`, which requires only either at most one remaining non-closed heavy vertex or $\mathrm{S1}_W$.

Step 1: Unified Phase E (`F3_star`). Suppose $Z \subseteq W$ has odd size, $h \in Z$, $P = \mathrm{pend}_W(Z,h)$ has $|P| = 3$, $X_0 = Z \cup P$ is an even good state with neutral dimension drop $\mathrm{cdim}_W(Z) = 2 + \mathrm{cdim}_W(X_0)$, and *at most one* member of P violates $(\star\star)_W$. (When $\mathrm{S1}_W$ holds, $P \subseteq \mathrm{rs}_W(h)$ cannot contain two distinct heavy vertices by $\mathrm{S1}_W$ applied at h itself; when $|H_{M,W}| \leq 2$, $h \in Z \cap H_{M,W}$ is already consumed, leaving at most one heavy vertex outside Z.) By recovery-triangle-freeness on h, P is independent in rs_W, and immediate closure (`BoundedNesting.imm_closure`) ensures $\mathrm{pend}_W(X_0,t) \neq \emptyset$ for all $t \in P$.

- If $|W \setminus X_0| = 1$, say $W \setminus X_0 = \{w\}$, then every $t \in P$ has $\mathrm{pend}_W(X_0,t) = \{w\}$; taking a test vector $x \in \mathrm{subCode}_M(W)$ vanishing on $Z \cup \{t_1\}$ with $x_{t_2} \neq 0$ (`BoundedNesting.test_vec`), row t_1 forces $x_w = 0$ and row t_2 then forces $x_{t_2} = 0$, contradiction.

- Suppose $|W \setminus X_0| \geq 3$ (recall $|W| - |X_0|$ is odd). Every light $t \in P$ has $|\mathrm{pend}_W(X_0,t)| \leq 2$ (and $\neq 1$ by Phase F1), hence $|\mathrm{pend}_W(X_0,t)| = 2$. If all three members of P are light, Phase F2 equates their 2-element pending sets to a common pair $\{a,b\}$, so $P \cup \bigcup_{t \in P} \mathrm{rs}_W(t) \subseteq P \cup \{h,a,b\}$ has size at most $6 = 2|P|$, violating Hall's count [18] on minimal counterexamples (`BoundedNesting.MinCE.hall_count`).

- Otherwise $P = \{u_1, u_2, g\}$ where g is the unique heavy member of P and u_1, u_2 are light. Both u_1 and u_2 have 2-element pending sets outside X_0, which coincide by Phase F2: $\mathrm{pend}_W(X_0,u_1) = \mathrm{pend}_W(X_0,u_2) = \{a,b\}$ $(a \neq b)$. Making the neutral move from X_0 along u_1 gives the exact drop $\mathrm{cdim}_W(X_0) = 1 + \mathrm{cdim}_W(X_0 \cup \{a,b\})$. If every $y \in \mathrm{subCode}_M(W) \cap \mathrm{zeroOn}(X_0)$ had $y_a = 0$, row u_1 ($M_{u_1,a}y_a + M_{u_1,b}y_b = 0$ with $M_{u_1,b} \neq 0$) would also force $y_b = 0$, contradicting $\mathrm{cdim}_W(X_0) > \mathrm{cdim}_W(X_0 \cup \{a,b\})$. Thus there exists $y \in \mathrm{subCode}_M(W) \cap \mathrm{zeroOn}(X_0)$ with $y_a \neq 0$ satisfying $M_{u_i,a}y_a + M_{u_i,b}y_b = 0$ for $i = 1,2$. Meanwhile, `BoundedNesting.test_vec` on (u_2, g) gives $x \in \mathrm{subCode}_M(W)$ vanishing on $Z \cup \{u_1\}$ with $x_{u_2} \neq 0$, so $M_{u_1,a}x_a + M_{u_1,b}x_b = 0$ while $x_{u_2} + M_{u_2,a}x_a + M_{u_2,b}x_b = 0$, contradicting proportional two-row elimination (`BoundedNesting.two_rows_zero`).

Step 2: Odd-state lemma and minimal counterexample exclusion (`odd_state_star`, `MinCE.false_SHStar`). Strong induction on the remaining non-closed heavy count $t = |\mathrm{NC}_W(X)|$ and $|W \setminus X|$ proves `BoundedNesting.OSAt`(M,W,v_0,t) for all t: an odd state $X \subsetneq W$ has no pending singleton, preserves parity under pending pairs, and on a pending triple $h \in X$ drops $|\mathrm{NC}_W|$ and hits contradiction by `F3_star`. Finally, on W itself: if some $v_0 \in W$ has $|\mathrm{rs}_W(v_0)| = 2$ (Case 1), $X_0 = \{v_0\} \cup \mathrm{rs}_W(v_0)$ is an odd state of size 3; if some heavy $h \in W$ has $|\mathrm{rs}_W(h)| = 3$ (Case 2), expanding $\{h\}$ invokes `F3_star` directly; otherwise (Case 3), every light vertex has degree 3, every heavy vertex has degree 4, and $(\star\star)_W$ plus $(\star 4)_W$ make rs_W a symmetric simple graph on W. Since $|W|$ is odd and the number of odd-degree (light) vertices in any simple graph is even, $|H_{M,W}|$ is odd (hence ≥ 1, and $= 1$ whenever $|H_{M,W}| \leq 2$, so $\mathrm{S1}_W$ holds unconditionally). Picking $h \in H_{M,W}$, $\mathrm{S1}_W$ at h allows at most one heavy neighbor among the four elements of $\mathrm{rs}_W(h)$, leaving at least three light neighbors $v_0 \in \mathrm{rs}_W(h)$. Starting at v_0, $\mathrm{S1}_W$ at v_0 guarantees that the other two neighbors $s_1, s_2 \in \mathrm{rs}_W(v_0) \setminus \{h\}$ are light, and the exact Case 3 closure of [7] contradicts `BoundedNesting.F1` or `odd_state_star`.

For (2) and (3), global instantiation $W = V$ gives `theoremSHStar`; transpose invariance $\dim_F \ker M^{\mathsf{T}} = \dim_F \ker M$ gives `theoremSHStar_transpose`; and on a triangle-free graph G

of maximum degree 4, `NoCommonNbhd` on H_M implies `BoundedNesting.SepS1`, so (2) bounds $\dim_F \ker M \le \lfloor |V|/2 \rfloor$ and rank-nullity $\operatorname{rank}_F M + \dim_F \ker M = |V|$ gives `theoremSHStar_graph`. □

4.2 Corollary M*: heavy-matching deficiency bound

For a simple graph $G = (V, E)$ and subset $H \subseteq V$, define the *common-neighbor-free packing number* (`rhoMatch`) by

$$\rho_{\mathrm{m}}(G, H) := \max\{|D| : D \subseteq H \text{ and } \forall u, v \in D,\ u \neq v \implies \texttt{NoCommonNbhd}(G, u, v)\}. \tag{2}$$

On a triangle-free graph G, two vertices $u \neq v$ satisfy $\texttt{NoCommonNbhd}(G, u, v)$ if and only if $\operatorname{dist}_G(u, v) \neq 2$, i.e., either $\{u, v\} \in E$ or $\operatorname{dist}_G(u, v) \ge 3$. Consequently, an admissible subset $D \subseteq H$ is any disjoint union of edges and isolated vertices in $G[H]$ whose connected components lie at pairwise G-distance at least 3. In particular, every distance-3 packing (`BoundedNesting.Far3`) is admissible, so $\rho_{\mathrm{m}}(G, H) \ge \rho_3(G, H)$ always holds, and whenever H spans an induced matching of r edges in G at pairwise distance at least 3, $\rho_{\mathrm{m}}(G, H) = 2r = 2\rho_3(G, H)$.

Corollary 4.2 (Barrier and matching bounds with ρ_{m})**.** Lean: `finrank_ker_le_barrier_SHStar`, `finrank_ker_add_card_le_matching_SHStar`, `rhoMatch`, `exists_rhoMatch`, `rho3_le_rhoMatch`, `finrank_ker_le_matching_rhoMatch`, `finrank_ker_le_matching_rhoMatch_transpose` [`TheoremSHStar.lean`]

Let F be a field and V be a finite vertex set.

1. (Barrier form.) *Suppose $M \in F^{V \times V}$ satisfies* `MotionDepthCodes.ThmSHyp`*, $U \subseteq V$, and $\mathcal{P}$ is a partition of $V \setminus U$ into pairwise disjoint blocks such that $M_{u,v} = 0$ between distinct blocks $B \neq B'$ of $\mathcal{P}$ and* $\texttt{SHStarOn}(M, B)$ *holds for every $B \in \mathcal{P}$. Then*
$$\dim_F \ker M \le |U| + \sum_{B \in \mathcal{P}} \lfloor |B|/2 \rfloor.$$

2. (Matching-packing form.) *Let $G = (V, E)$ be a triangle-free simple graph of maximum degree at most 4 with matching number $\nu(G)$, and suppose $M \in F^{V \times V}$ has unit diagonal and off-diagonal support in G. Then $\rho_3(G, H_M) \le \rho_{\mathrm{m}}(G, H_M)$ and*
$$\dim_F \ker M + \max\big(\rho_{\mathrm{m}}(G, H_M),\, \min(|H_M|, 2)\big) \le \nu(G) + |H_M|,$$
and the same inequality holds with $H_{M^{\mathsf{T}}}$ in place of H_M.

Outline of the formal proof. For (1) (`finrank_ker_le_barrier_SHStar`), intersecting $\ker M$ with $\operatorname{zeroOn}(U)$ loses at most $|U|$ dimensions and embeds into $\bigoplus_{B \in \mathcal{P}} \operatorname{subCode}_M(B)$ because M has no cross-block entries; applying Theorem 4.1(1) to each block $B \in \mathcal{P}$ gives $\sum_{B \in \mathcal{P}} \lfloor |B|/2 \rfloor$.

For (2) (`finrank_ker_add_card_le_matching_SHStar` and `finrank_ker_le_matching_rhoMatch`), let $D \subseteq H_M$ satisfy either $|D| \le 2$ or pairwise `NoCommonNbhd` on (G, u, v), and set $U = H_M \setminus D$. By the Tutte–Berge formula [5] (Lean-verified in `lean-machine` as `SimpleGraph.exists_card_add_sum_card_div_two_eq_matchingNumber`), there is a Tutte barrier $U' \subseteq V$ such that $|U'| + \sum_{B \in \operatorname{comp}(G - U')} \lfloor |B|/2 \rfloor = \nu(G)$. On each punctured component $B' = B \setminus U$ of $V \setminus (U \cup U')$, every heavy vertex of M in B' belongs to D, so `SHStarOn` holds on (M, B'). Applying (1) with cut $U \cup U'$ gives
$$\dim_F \ker M \le |U \cup U'| + \sum_{B} \lfloor |B \setminus U|/2 \rfloor \le |U| + \nu(G) = |H_M| - |D| + \nu(G).$$
Choosing D to attain $\rho_{\mathrm{m}}(G, H_M)$ (`exists_rhoMatch`) or $\min(|H_M|, 2)$ completes the proof, and transpose invariance of $\dim_F \ker M$ gives `finrank_ker_le_matching_rhoMatch_transpose`. □

Remark 4.3 (Linear-size separation between ρ_{m} and ρ_3)**.** For each integer $k \geq 2$, construct a connected bipartite (hence triangle-free) graph G_{8k} on $8k$ vertices with bipartition $A \sqcup B$, where $A = \bigsqcup_{r=0}^{k-1}\{a_{r,0}, a_{r,1}, a_{r,2}, a_{r,3}\}$ and $B = \bigsqcup_{r=0}^{k-1}\{b_{r,0}, b_{r,1}, b_{r,2}, b_{r,3}\}$, by including, for each $0 \leq r < k$, all seven edges incident to $a_{r,0}$ or $b_{r,0}$ inside block r ($\{a_{r,0}, b_{r,j}\}$ and $\{a_{r,j}, b_{r,0}\}$ for $0 \leq j \leq 3$) together with a 2-regular bipartite subgraph on the remaining $6k$ vertices $\{a_{r,j}, b_{r,j} : 0 \leq r < k,\ 1 \leq j \leq 3\}$ having edges $\{a_{r,j}, b_{r,j}\}$ and $\{a_{r,j}, b_{(r+1) \bmod k,\,(j \bmod 3)+1}\}$ (a single $6k$-cycle when $3 \nmid k$, and three disjoint $2k$-cycles when $3 \mid k$). Then G_{8k} is 4-regular on $H = \{a_{r,0}, b_{r,0} : 0 \leq r < k\}$ ($|H| = 2k$), 3-regular on $V \setminus H$, and has a perfect matching $\nu(G_{8k}) = 4k$ (verified numerically for $2 \leq k \leq 6$ by `tools/reproduce_numerics.py`). Because G_{8k} is bipartite and the neighborhoods $N(a_{r,0}) = \{b_{r,0}, \ldots, b_{r,3}\}$ and $N(b_{r,0}) = \{a_{r,0}, \ldots, a_{r,3}\}$ are pairwise disjoint across r, no two distinct vertices of H share a common neighbor in G_{8k}, whereas each pair $\{a_{r,0}, b_{r,0}\}$ is an edge:

- the eighth paper's distance-3 packing invariant satisfies $\rho_3(G_{8k}, H) = k$, yielding only the upper bound $\dim_F \ker M \leq 4k + 2k - k = 5k = \frac{5}{8}|V|$;
- by contrast, $\rho_{\mathrm{m}}(G_{8k}, H) = 2k = |H|$, so Theorem 4.1 and Corollary 4.2 certify the sharp half-rate bound $\dim_F \ker M \leq 4k = \frac{1}{2}|V|$ over every field F; over $\mathbb{F}_2$, the all-ones unit-diagonal matrix supported on G_{8k} has heavy set H and attains $\dim_{\mathbb{F}_2} \ker M = 4k$ (verified for $2 \leq k \leq 6$ by `tools/reproduce_numerics.py`).

5 Deep-Nesting Stroke Laws, Super-Fibonacci Unwrapping, and Block Pairs

While Sections 3 and 4 push half-rate coding barriers across $\mathcal{S}$ and degree-4 graphs under (S1), completing the conditional impossibility program of [8, 7] for 2-head tapes also requires understanding when 2-head walk dependency DAGs admit εT-edge cuts to bounded depth (TCP(2)). The canonical obstruction testbed recalled in [7, Section 2.5] (after [8, Definitions 5.3 and 5.9]) is the *deep-nesting pair* $\mathrm{DNP}(s_0, m)$, running two copies of the recursive turn-and-revisit walk $U(s_0, m, 1)$ at relative time shift $\xi \in \{0, \ldots, P-1\}$ over a shared tape (Section 2). On a single copy of $U(s_0, m, 1)$, a turn-supported label $\Phi : \mathbb{Z} \to \mathbb{Z}_q$ cuts depth to $2^{m+O(1)}$ by keeping $\Phi(t+1) - \Phi(t) \equiv 1 \pmod q$ on plain steps and adjusting Φ at turns so that same-cell revisit hops satisfy $\Phi(u) - \Phi(p) \equiv 1 \pmod q$. On two shifted copies, however, a revisit hop (p, u) of head 1 is *interrupted* (non-blind) whenever head 2 executes a turn in $(p, u]$, adding head 2's net potential increment F across $(p, u]$ to the mod-q consistency relation.

In [7, Section 5], the path-sum law, band decomposition, block-label family $B(a, K)$, and straddle involution were developed on paper without Lean formalization, and Conjecture A″ [7, Section 5.6, Conjecture 5.24]—asserting that at some odd shift ξ, every mod-q label of cost at most $n_{\mathrm{h}}/8$ on the r-fold cover has at most $C \log_2 q + C' + \eta(m+1)$ doubly θ_0-served levels ($\eta < 1$)—left two explicit steps open at the end of [7, Section 5.6] even for pairs of block labels:

1. *"a bound on $|F|$ on the straddling hops, or a version of Theorem V that counts the window directly"* (since $|F|$ in [7, Theorem 5.21] can grow linearly in m and wrap modulo q unless q is of order $(m - \kappa)Q$); and
2. *"at one fixed shift for all coarse classes, equidistribution of a class's apex centres relative to the other head's canceller valleys of level about $k + \log_2(m - k)$, a mixed-radix carry problem".*

In this section we formalize the turn-potential, V-straddle, and super-Fibonacci apparatus in Lean 4: Theorem 5.2 bypasses step (1) altogether for level-uniform labels by summing straddling pairs

directly modulo q with zero wrap-around loss, `exists_oddShift_coarse_straddle` establishes a weaker one-direction $\frac{1}{16}$-fraction form of step (2) over odd shifts, and Theorem 5.5 combines these with Theorem 5.3 to produce three-regime failure witnesses for all block-label pairs $(B(a_1, K_1), B(a_2, K_2))$ of bounded block length at two fixed odd shifts (a single shift for symmetric pairs).

5.1 Exact stroke law and $O(m)$ recursive turn potential

Fix integers $0 \le m \le s_0$, dyadic scales $R_i = 2^{s_0 - i}$ (`scaleR`), stroke lengths $\Pi_\kappa = (m - \kappa + 1)R_\kappa$ (`strokeLen`), and fundamental period $P = 4\Pi_0 = 4(m+1)2^{s_0}$ (`dnpPeriod`) of $U(s_0, m, 1)$ [8, Definition 5.3], [7, Section 2.5]. A *level-uniform turn-supported potential* on $U(s_0, m, 1)$ is specified by integer level parameters $A, E : \mathbb{N} \to \mathbb{Z}$ (or equivalently back-stroke and forward-stroke path-sum increments $c_{\mathrm{b}}(\kappa) = 1 + A_\kappa$ and $c_{\mathrm{f}}(\kappa) = 2 + A_\kappa + E_\kappa$, `pathSumBack` and `pathSumFwd`) together with an outer tip jump $j_{\mathrm{tip}} \in \mathbb{Z}$ at the period reversals Π_0 and $3\Pi_0$: at every internal apex of level κ $(1 \le \kappa \le m)$, the turn jump adds $c_{\mathrm{b}}(\kappa)$; at every internal valley of level κ, the turn jump adds $c_{\mathrm{f}}(\kappa) - c_{\mathrm{b}}(\kappa) - M_{\mathrm{f}}(\kappa)$, where $M_{\mathrm{f}}(\kappa) = \sum_{l=\kappa+1}^{m} c_{\mathrm{f}}(l)$ (`strokeMass`) is the cumulative turn jump inside one level-κ stroke, arranged so that entering the level-κ forward stroke starts at cumulative offset $c_{\mathrm{f}}(\kappa)$ and completing the entire level-κ V contributes net turn jump $c_{\mathrm{f}}(\kappa) + M_{\mathrm{f}}(\kappa) = M_{\mathrm{f}}(\kappa - 1)$.

Proposition 5.1 (Exact stroke law and recursive turn potential)**.** Lean: `scaleR`, `strokeLen`, `dnpPeriod`, `decompDepth_valid`, `reconstruct_decomp`, `decomp_reconstruct`, `card_stroke_stopDepth_branchWord`, `card_stroke_stopDepth`, `card_stroke_stopDepth_branchWord_parity`, `card_stroke_stopDepth_parity`, `strokePhiRec`, `strokePhiRec_back`, `strokePhiRec_fwd`, `evalPhiRec`, `evalPhiRec_eq_sum_turnJumps` [DNPStrokes.lean]

Let $0 \le \kappa \le m \le s_0$.

1. (Stroke decomposition bijection.) *Recursive stroke decomposition (*`decompDepth`*) and reconstruction (*`reconstructDepth`*) establish an explicit bijection between offsets* $o \in \{0, \dots, \Pi_\kappa - 1\}$ *of a level-*κ *stroke and valid triples* (s, w, rem) *(*`ValidStrokeCode`*) consisting of a stopping depth* $s \in \{0, \dots, m - \kappa\}$*, a nested branch word* $w \in \{\mathrm{b}, \mathrm{f}\}^s$ *(*false $=$ b, true $=$ f*), and a plain-segment remainder* $\mathrm{rem} \in \{0, \dots, R_{\kappa+s} - 1\}$.

2. (Uniform stopping depth, fair-coin branches, and parity independence.) *For each* $0 \le s \le m - \kappa$ *and each word* $w \in \{\mathrm{b}, \mathrm{f}\}^s$*, there are exactly* $R_{\kappa+s} = R_\kappa / 2^s$ *offsets* $o \in [0, \Pi_\kappa)$ *with code* $(s, w, \cdot)$*, and summing over* $w \in \{\mathrm{b}, \mathrm{f}\}^s$ *gives* exactly $R_\kappa = \Pi_\kappa / (m - \kappa + 1)$ *offsets at stopping depth* s*. Moreover, when* $m < s_0$ *(so every* R_i *for* $i \le m$ *is even),* $o \equiv \mathrm{rem} \pmod 2$*, and for each parity* $p \in \{0, 1\}$*, stopping depth* s*, and word* $w \in \{\mathrm{b}, \mathrm{f}\}^s$*, the joint count is* $R_{\kappa+s}/2$ *and the marginal count at* (s, p) *is* $R_\kappa / 2$.

3. ($O(m)$ recursive turn potential and V-stroke decomposition.) *For any* $c_{\mathrm{b}}, c_{\mathrm{f}} : \mathbb{N} \to \mathbb{Z}$ *and* $j_{\mathrm{tip}} \in \mathbb{Z}$*, let* $Y_i(o) =$ `strokePhiRec`$(s_0, m, c_{\mathrm{b}}, c_{\mathrm{f}}, i, o)$ *be the* $O(m - i)$ *recursive stroke potential at level* $i \le m$*. The* $O(m)$*-step period evaluator* $\Phi(t) =$ `evalPhiRec`$(s_0, m, c_{\mathrm{b}}, c_{\mathrm{f}}, j_{\mathrm{tip}}, t)$ *equals the exact cumulative sum of turn jumps* $\sum_{0 < \tau \le t} J(\tau)$ *(*`periodTurnJump`*) for all* $0 \le t < P$*. Inside any enclosing level-*$(\kappa - 1)$ *stroke* $(1 \le \kappa \le m)$*, its internal level-*κ *V at apex offset* $R_{\kappa-1}$ *satisfies, for all* $0 \le o < \Pi_\kappa$*,*

$$Y_{\kappa-1}(R_{\kappa-1} + o) = c_{\mathrm{b}}(\kappa) + Y_\kappa(o), \qquad Y_{\kappa-1}(R_{\kappa-1} + \Pi_\kappa + o) = c_{\mathrm{f}}(\kappa) + Y_\kappa(o). \tag{3}$$

Outline of the formal proof. Part (1) (`decompDepth_valid`, `reconstruct_decomp`, `decomp_reconstruct`) inducts on remaining depth $d = m - \kappa$ using the stroke-length identity

$\Pi_\kappa = R_\kappa + 2\Pi_{\kappa+1}$ (`strokeLen_succ`), partitioning $[0, \Pi_\kappa)$ into the initial plain segment $[0, R_\kappa)$ ($s = 0$), the back sub-stroke $[R_\kappa, R_\kappa + \Pi_{\kappa+1})$ ($b_1 = \mathrm{b}$), and the forward sub-stroke $[R_\kappa + \Pi_{\kappa+1}, \Pi_\kappa)$ ($b_1 = \mathrm{f}$). Part (2) (`card_stroke_stopDepth_branchWord`, `card_stroke_stopDepth`, `card_stroke_stopDepth_branchWord_parity`, and `card_stroke_stopDepth_parity`) counts fibers via `Finset.card_bij`; when $m < s_0$, $2 \mid R_m$ divides every summand R_i and Π_i added by `reconstructDepth`, so $o \bmod 2 = \mathrm{rem} \bmod 2$ and $[0, R_{\kappa+s})$ splits into $R_{\kappa+s}/2$ even and $R_{\kappa+s}/2$ odd remainders. For (3), unfolding one step of `strokePhiRec` yields (3) (`strokePhiRec_back`, `strokePhiRec_fwd`), induction on $m - \kappa$ equates `strokePhiRec` with the sum of `strokeTurnJump` over $(0, o]$ (`strokePhiRec_eq_sum_turnJumps`) by cancelling $-M_\mathrm{f}(\kappa)$ at the valley turn against the completed back-stroke jump sum $M_\mathrm{f}(\kappa)$, and summing across the four level-0 strokes of $[0, P)$ establishes `evalPhiRec_eq_sum_turnJumps`. □

5.2 Unconditional mod-q V-straddle bound and odd-center straddle sums

Consider a coarse revisit hop $(c_0 - d, c_0 + d)$ of half-width $d \in \{1, \ldots, R_k - 1\}$ centered at an apex mapped to time $c_0 \in \mathbb{Z}$ on the other head's timeline, and suppose the other head executes a level-κ V on $[\alpha_V, \alpha_V + 2\Pi_\kappa)$ with $\Pi_\kappa \le R_k$. We say d *straddles* this V (`straddleSet`) if its left endpoint lands in the back stroke and its right endpoint lands in the forward stroke:

$$d \in \mathrm{Strad}(c_0, \alpha_V, \Pi_\kappa, R_k) \iff 1 \le d \le R_k - 1, \quad \alpha_V \le c_0 - d < \alpha_V + \Pi_\kappa \le c_0 + d < \alpha_V + 2\Pi_\kappa. \tag{4}$$

Notice the reflection symmetry $d \mapsto \Pi_\kappa - d$ (`mem_straddleSet_sub_iff`): when $\Pi_\kappa \le R_k$, $d \in \mathrm{Strad}(c_0, \alpha_V, \Pi_\kappa, R_k)$ if and only if $\Pi_\kappa - d \in \mathrm{Strad}(c_0, \alpha_V, \Pi_\kappa, R_k)$, because the back-stroke offset $o_p(d) = (c_0 - d) - \alpha_V$ and forward-stroke offset $o_u(d) = (c_0 + d) - (\alpha_V + \Pi_\kappa)$ swap under $d \leftrightarrow \Pi_\kappa - d$:

$$o_p(\Pi_\kappa - d) = (c_0 - \Pi_\kappa + d) - \alpha_V = o_u(d), \qquad o_u(\Pi_\kappa - d) = (c_0 + \Pi_\kappa - d) - (\alpha_V + \Pi_\kappa) = o_p(d).$$

Theorem 5.2 (Unconditional mod-q V-straddle inequality and parity-invariant straddle count)**.** Lean: `straddleSet`, `mem_straddleSet_sub_iff`, `HasLevelUniformVDecomp`, `strokePhiRec_hasLevelUniformVDecomp`, `straddle_otherHeadInc_sum`, `theoremVStar_pair`, `theoremVStar_sum`, `sum_straddleSet_card_parity` [`DNPStraddle.lean`]

Let $\Phi : \mathbb{Z} \to \mathbb{Z}$ admit a level-uniform V decomposition on $[\alpha_V, \alpha_V + 2\Pi_\kappa)$ with base constant $C_V \in \mathbb{Z}$ and parameters $(c_\mathrm{b}, c_\mathrm{f}, Y)$, i.e., $\Phi(\alpha_V + o) = C_V + c_\mathrm{b} + Y(o)$ and $\Phi(\alpha_V + \Pi_\kappa + o) = C_V + c_\mathrm{f} + Y(o)$ for $0 \le o < \Pi_\kappa$ (`HasLevelUniformVDecomp`; instantiated for `strokePhiRec` via (3) by `strokePhiRec_hasLevelUniformVDecomp`), and write $F(d) = \Phi(c_0 + d) - \Phi(c_0 - d)$ (`otherHeadInc`) for the other head's net potential increment across $(c_0 - d, c_0 + d)$.

1. (Exact $\mathbb{Z}$-cancellation.) *For every $d \in \mathrm{Strad}(c_0, \alpha_V, \Pi_\kappa, R_k)$,*

$$F(d) + F(\Pi_\kappa - d) = 2(c_\mathrm{f} - c_\mathrm{b}) \quad \text{in } \mathbb{Z}.$$

2. (Unconditional mod-q pair and set bounds.) *Let $q > 0$ and $A' \in \mathbb{Z}$, and define the reduced mod-q interrupted hop error $e(d) = (A' + F(d)) \bmod q \in \{0, \ldots, q - 1\}$. For every $d \in S := \mathrm{Strad}(c_0, \alpha_V, \Pi_\kappa, R_k)$,*

$$(2A' + 2(c_\mathrm{f} - c_\mathrm{b})) \bmod q \;\le\; e(d) + e(\Pi_\kappa - d), \tag{5}$$

and whenever $\Pi_\kappa \le R_k$, summing over $d \in S$ yields

$$|S| \cdot ((2A' + 2(c_\mathrm{f} - c_\mathrm{b})) \bmod q) \;\le\; 2 \sum_{d \in S} e(d), \tag{6}$$

with zero dependence on m or on the integer magnitude of Y or $F(d)$.

3. (Exact parity-conditioned straddle count.) *Suppose* $\Pi_\kappa = 2L$ *is even and* $2L \le R_k$. *For* either *fixed parity* $p \in \{0,1\}$, *summing* $|\mathrm{Strad}(c_0, \alpha_V, 2L, R_k)|$ *over all centers* $c_0 \in [\alpha_V, \alpha_V + 4L)$ *with* $c_0 \bmod 2 = p$ *equals* exactly $L^2 = \Pi_\kappa^2/4$.

Outline of the formal proof. For (1) (`straddle_otherHeadInc_sum`), the V decomposition gives $F(d) = (c_\mathrm{f} - c_\mathrm{b}) + Y(o_u(d)) - Y(o_p(d))$ and $F(\Pi_\kappa - d) = (c_\mathrm{f} - c_\mathrm{b}) + Y(o_p(d)) - Y(o_u(d))$; adding cancels $Y(o_u(d)) - Y(o_p(d))$ identically in $\mathbb{Z}$.

For (2) (`theoremVStar_pair`, `theoremVStar_sum`), $e(d) + e(\Pi_\kappa - d) \ge 0$ is congruent modulo q to $(A' + F(d)) + (A' + F(\Pi_\kappa - d)) = 2A' + 2(c_\mathrm{f} - c_\mathrm{b})$; since for any integer $S_\mathrm{sum} \ge 0$ and $q > 0$ we always have $S_\mathrm{sum} \bmod q \le S_\mathrm{sum}$, (5) holds without any wrap-around loss. Because $d \mapsto \Pi_\kappa - d$ is an involution of S when $\Pi_\kappa \le R_k$, $\sum_{d\in S}(e(d) + e(\Pi_\kappa - d)) = 2\sum_{d\in S} e(d)$, giving (6).

For (3) (`sum_straddleSet_card_parity`), swap the order of summation between $c_0 \in [\alpha_V, \alpha_V + 4L)$ ($c_0 \bmod 2 = p$) and $d \in \{1, \dots, 2L-1\}$. For each fixed $d \in \{1, \dots, 2L-1\}$, solving the inequalities in (4) for c_0 shows that $\{c_0 : d \in \mathrm{Strad}(c_0, \alpha_V, 2L, R_k)\}$ is the half-open interval $[\alpha_V + \max(d, 2L-d),\ \alpha_V + \max(d, 2L-d) + 2\min(d, 2L-d))$, whose length $2\min(d, 2L-d)$ is even. Every half-open integer interval $[a, a+2M)$ of even length contains *exactly* M even integers and *exactly* M odd integers (`card_Ico_even_len_mod_two`), so the parity-p fiber has exact size $\min(d, 2L-d)$, and $\sum_{d=1}^{2L-1} \min(d, 2L-d) = L^2$ (`sum_min_sub_two_mul`). □

5.3 Super-Fibonacci unwrapping for level-uniform labels

On a single head of $U(s_0, m, 1)$ carrying a level-uniform potential (A, E) modulo $q \ge 3$, consider the blind (uninterrupted) revisit hops around the apices and valleys of level k:

- every blind apex hop at level k has reduced mod-q hop error $a(k) := A_k \bmod q \in [0, q)$ (`blindApexErr`, verified on `strokePhiRec` by `strokePhiRec_apex_hop`);
- for $1 \le k \le l \le m$, every blind valley hop at level k whose half-width stops in the plain segment of nested level l (*valley band* $l - k$) has reduced mod-q hop error (`strokePhiRec_valley_band_hop`)

$$b_{l-k}(k) = \texttt{blindBandErr}(A, E, q, k, l) := \Big(E_k - \sum_{i=k+1}^{l} (A_i + E_i + 2)\Big) \bmod q \in [0, q). \quad (7)$$

Whenever a level-k class is θ-served ($\theta \le 1/256$) and at least half of its apex hop family or valley sub-band $0, 1, 2$ is blind at the chosen shift (by [7, Proposition 5.17 and Corollary 5.18], fine classes are mostly blind on average over shifts, whereas at a fixed shift half-blindness is an explicit hypothesis), Markov's inequality bounds each corresponding reduced blind error by $q/8$ (`blind_error_le_div_eight`). Unwrapping (7) across bands $0, 1, 2$ forces *super-Fibonacci* growth of the blind errors from fine to coarse scales, unconditionally ruling out $s \ge 2\lfloor \log_2 q \rfloor + 3$ consecutive levels of blind errors at most $q/8$ (and hence bounding consecutive half-blind θ-served levels conditionally on half-blindness at the chosen shift).

Theorem 5.3 (Super-Fibonacci unwrapping for level-uniform labels)**.** Lean: `strokePhiRec_apex_hop`, `strokePhiRec_valley_band_hop`, `blindApexErr`, `blindBandErr`, `unwrap_blindBand_step`, `superFib_step`, `superFib_lower_bound`, `log_bound_of_fib_le`, `blind_error_le_div_eight`, `theoremLU_consecutive` [DNPFibonacci.lean]

Let $A, E : \mathbb{N} \to \mathbb{Z}$ *and* $q \ge 3$.

1. (Exact $\mathbb{Z}$-unwrapping and super-Fibonacci step.) *For any $k \le l$, if* $\texttt{blindBandErr}(A, E, q, k, l+1)$, $\texttt{blindApexErr}(A, q, l+1)$, *and* $\texttt{blindBandErr}(A, E, q, l+1, l+1)$ *are each at most $\lfloor q/8 \rfloor$ (i.e., $8e \le q$), then the mod-q relation lifts to exact equality in $\mathbb{Z}$:*

$$\begin{aligned}&\texttt{blindBandErr}(A, E, q, k, l) - \texttt{blindBandErr}(A, E, q, k, l+1)\\ &\quad = \texttt{blindApexErr}(A, q, l+1) + \texttt{blindBandErr}(A, E, q, l+1, l+1) + 2.\end{aligned}$$

Consequently, if $b_1(k), b_2(k), a(k+1), b_0(k+1), a(k+2), b_0(k+2)$ are each at most $q/8$, adding the (k, k) and $(k, k+1)$ identities telescopes $b_1(k)$ and yields the super-Fibonacci inequality

$$b_0(k+1) + b_0(k+2) + 4 \;\le\; b_0(k). \tag{8}$$

2. (Fibonacci lower bound and $2\lfloor \log_2 q \rfloor + 2$ run limit.) *Suppose $s \ge 1$ consecutive levels $k_0, \dots, k_0 + s - 1$ satisfy $8a(k_0 + i) \le q$ and $8b_0(k_0 + i) \le q$ for $0 \le i < s$, $8b_1(k_0 + i) \le q$ for $i + 1 < s$, and $8b_2(k_0 + i) \le q$ for $i + 2 < s$. Writing $F_s = \texttt{Nat.fib}(s)$ ($F_0 = 0, F_1 = F_2 = 1, F_3 = 2, \dots$), we have*

$$4(F_s - 1) \le b_0(k_0), \qquad 32(F_s - 1) \le q, \qquad \text{and} \qquad s \le 2\lfloor \log_2 q \rfloor + 2.$$

Outline of the formal proof. For (1) (`unwrap_blindBand_step`), write ex, ey, ea, and ee for `blindBandErr` at (A, E, q, k, l), `blindBandErr` at $(A, E, q, k, l+1)$, `blindApexErr` at $(A, q, l+1)$, and `blindBandErr` at $(A, E, q, l+1, l+1)$, and let $D = (\text{ex} - \text{ey}) - (\text{ea} + \text{ee} + 2)$. By (7), $q \mid D$. Nonnegativity gives $D \le \text{ex} - 2 < q$. For the lower bound, if $q \ge 8$, the hypotheses $8 \cdot \text{ey} \le q$, $8 \cdot \text{ea} \le q$, and $8 \cdot \text{ee} \le q$ give $\text{ey} + \text{ea} + \text{ee} + 2 \le 3q/8 + 2 < q$, while if $3 \le q < 8$, $8x \le q$ forces $\text{ey} = \text{ea} = \text{ee} = 0$ and hence $D = \text{ex} - 2 \ge -2 > -q$. Thus $|D| < q$ and $q \mid D$, forcing $D = 0$. Applying the step lemma `unwrap_blindBand_step` at $l = k$ (using $8b_1(k) \le q$) and at $l = k + 1$ (using $8b_2(k) \le q$) sums to $b_0(k) - b_2(k) = a(k+1) + b_0(k+1) + a(k+2) + b_0(k+2) + 4$; dropping the nonnegative terms $b_2(k), a(k+1), a(k+2) \ge 0$ yields (8) (`superFib_step`).

For (2) (`superFib_lower_bound`, `log_bound_of_fib_le`, `theoremLU_consecutive`), strong induction on s shows $b_0(k_0) \ge 4(F_s - 1)$ since $4(F_{s-1} - 1) + 4(F_{s-2} - 1) + 4 = 4(F_s - 1)$. Combining with $8b_0(k_0) \le q$ gives $32(F_s - 1) \le q$. Using $F_{2k+3} \ge 2^k + 1$ (`two_pow_add_one_le_fib`) for $s \ge 3$ ($k = \lfloor (s-3)/2 \rfloor$) gives $2^{k+5} \le 32(F_s - 1) \le q$, so $k + 5 \le \lfloor \log_2 q \rfloor$ and $s \le 2k + 4 \le 2\lfloor \log_2 q \rfloor - 6 < 2\lfloor \log_2 q \rfloor + 2$. □

5.4 Block-cancellation drift, Mersenne clocks, and bounded-range block pairs

To evade Theorem 5.3 over more than $O(\log q)$ levels on a single walk, [7, Section 5.3] introduced the periodic *block-cancellation family* $B(a, K)$ ($a \ge 1$, $K \in \mathbb{Z}$): partition $\{0, \dots, m\}$ from the bottom level m upward into blocks of length $a + 1$, designating κ as a *canceller* if $(m - \kappa) \bmod (a + 1) = a$ (`isCanceller`) and as a *served* level with $j = (m - \kappa) \bmod (a + 1) \in \{0, \dots, a - 1\}$ served levels below it otherwise. Setting $Q(a, K) = (2^a - 1)(2K + 2)$ (`blockQ`), define (`blockA`, `blockE`):

$$A_\kappa = \begin{cases} 0 & \text{if } \kappa \text{ is a canceller,} \\ K & \text{if } \kappa \text{ is served,} \end{cases} \qquad E_\kappa = \begin{cases} -Q(a, K) - 2 & \text{if } \kappa \text{ is a canceller,} \\ K + (2^j - 1)(2K + 2) & \text{if } \kappa \text{ is served.} \end{cases}$$

On served levels (`block_served_mass`), $A_\kappa + E_\kappa + 2 = 2^j(2K + 2)$, $c_{\mathrm{b}}(\kappa) = 1 + K$, and $c_{\mathrm{f}}(\kappa) = 2^j(2K+2)$; on cancellers (`block_canceller_mass`), $c_{\mathrm{b}}(\kappa) = 1$, $c_{\mathrm{f}}(\kappa) = -Q(a, K)$, and $c_{\mathrm{f}}(\kappa) - c_{\mathrm{b}}(\kappa) = -Q(a, K) - 1$, resetting the cumulative sum $\sum(A_i + E_i + 2)$ across each $(a + 1)$-level block to 0.

Moreover, reducing $K \mapsto K \bmod q$ leaves every mod-q blind error invariant (`block_blind_errors_mod_K`).

We first record the exact path-sum drift laws of $B(a, K)$ and the Mersenne-modulus behavior of the pure binary clock $B(\infty, 0)$ ($c_{\mathrm{b}} = 1$, $c_{\mathrm{f}}(\kappa) = 2^{m-\kappa+1}$, $A \equiv 0$, which makes every single-head hop error 0 in $\mathbb{Z}$ before modular reduction, and modulo a Mersenne number $q = 2^B - 1$ makes single-head hop errors vanish periodically in B, yet incurs straddle residue $q - 1$):

Proposition 5.4 (One-sided block drift and Mersenne straddle failure)**.** Lean: `blockQ`, `blockA`, `blockE`, `block_served_mass`, `block_canceller_mass`, `block_blind_errors_mod_K`, `propOC_fwd_bounds`, `propOC_back_bounds`, `propOC_sum_all_paths`, `propOC_mersenne_straddle` [DNPConjectureApp.lean]

Let $a \geq 1$ and $K \geq 0$, and for branch choices $(d_{\mathrm{canc}}, d_{\mathrm{srv}}) \in \{\mathrm{b}, \mathrm{f}\} \times \{\mathrm{b}, \mathrm{f}\}^a$ across one complete $(a+1)$-level block define the block path sum $\texttt{blockPathSum}(a, K, d_{\mathrm{canc}}, d_{\mathrm{srv}})$.

1. (Forward vs. back canceller bounds.) *When $d_{\mathrm{canc}} = \mathrm{f}$, the block path sum lies in $[-Q(a, K) + a(1 + K), 0]$ and equals 0 when $d_{\mathrm{srv}} \equiv \mathrm{f}$. When $d_{\mathrm{canc}} = \mathrm{b}$, the block path sum lies in $[1 + a(1 + K), Q(a, K) + 1]$.*

2. (Positive mean block drift.) *For every $K \in \mathbb{Z}$, summing* $\texttt{blockPathSum}(a, K, p_1, p_2)$ *over all 2^{a+1} branch choices $p \in \{\mathrm{b}, \mathrm{f}\} \times \{\mathrm{b}, \mathrm{f}\}^a$ equals $2^a(a(K + 1) + 1)$, which is strictly positive for all $K \geq 0$ (mean drift $+\frac{1}{2}(a(K + 1) + 1)$ per $(a + 1)$-level block under the fair-coin law of Proposition 5.1(2)).*

3. (Mersenne binary-clock straddle residue.) *For any $B \geq 2$ and Mersenne modulus $q = 2^B - 1$, the pure binary clock ($c_{\mathrm{b}} = 1$, $c_{\mathrm{f}}(\kappa) = 2^{m-\kappa+1}$, $A' = 0$) satisfies, at every level $\kappa \leq m$ with $B \mid (m - \kappa + 2)$,*

$$(2A' + 2(c_{\mathrm{f}}(\kappa) - c_{\mathrm{b}}(\kappa))) \bmod q = q - 1.$$

Outline of the formal proof. Parts (1) and (2) (`propOC_fwd_bounds`, `propOC_back_bounds`, and `propOC_sum_all_paths`, proved in Lean for all $a \geq 0$) use the geometric identity $\sum_{j=0}^{a-1} 2^j = 2^a - 1$ (`sum_fin_two_pow`) and coordinatewise involution $f_j \mapsto \neg f_j$ on $\{\mathrm{b}, \mathrm{f}\}^a$: summing $d_{\mathrm{canc}} \in \{\mathrm{b}, \mathrm{f}\}$ gives $1 - Q(a, K)$, while each served coordinate j averages $\frac{1}{2}(2^j(2K + 2) + 1 + K)$, summing to $Q(a, K) + a(1 + K)$ and leaving net mean $\frac{1}{2}(1 + a(1 + K))$. For (3) (`propOC_mersenne_straddle`), $B \mid (m - \kappa + 2)$ implies $(2^B - 1) \mid (2^{m-\kappa+2} - 1)$, so $2(2^{m-\kappa+1} - 1) = (2^{m-\kappa+2} - 1) - 1 \equiv -1 \equiv q - 1 \pmod q$. □

We now assemble Proposition 5.1, Theorem 5.2, and Theorem 5.3 with prefix lockstep and simultaneous reverse-Markov shift selection to establish three-regime failure witnesses on $\mathrm{DNP}(s_0, m)$ for all block-label pairs $(B(a_1, K_1), B(a_2, K_2))$ of bounded block length. For $1 \leq k \leq \kappa \leq m$, write $\texttt{periodVApices}(s_0, m, \kappa) \subseteq [0, P)$ for the set of $2^{\kappa+1}$ level-κ V start times in one period, $\texttt{oddRange}(P) = \{\xi \in [0, P) : \xi \bmod 2 = 1\}$, $\texttt{cancellerBelow}(m, a, k)$ for the first canceller level $\kappa \in [k + \lfloor \log_2(m + 1) \rfloor + 1, k + \lfloor \log_2(m + 1) \rfloor + 1 + a]$ of $B(a, \cdot)$ satisfying $\Pi_\kappa \leq R_k$, and $\texttt{classStraddleCount}(s_0, m, k, \kappa, \mathrm{dir}, \xi)$ for the total number of straddling half-widths d across all 2^{k+1} apices of level k on the active head (dir = true for head 1 at $(c + \xi) \bmod P$, dir = false for head 2 at $(c + P - \xi) \bmod P$) against all level-κ V's of the other head.

Theorem 5.5 (Three-regime witnesses for bounded-range block pairs on DNP)**.** Lean: `apexOffset_diff_indep_prefix`, `periodVApices_spans_disjoint`, `card_periodVApices`, `even_of_mem_periodVApices`, `sum_classStraddleCount_odd`, `reverse_markov_simultaneous`, `classStraddleCount`, `cancellerBelow_props`, `exists_oddShift_coarse_straddle`, `block_straddle_residue_ge_half_q`, `block_overflow_band_witness`, `theoremA_three_regimes`, `theoremA_symmetric_blockPair` [DNPConjectureApp.lean]

Let $a_1, a_2 \geq 1$, $m < s_0$, and $4(\lfloor \log_2(m+1) \rfloor + 1 + \max(a_1, a_2)) \leq m+1$. Set $\Omega = \texttt{oddRange}(P)$ and $K_{\text{coarse}} = \{1, \ldots, \lfloor (m+1)/4 \rfloor\}$. There exist two fixed odd shifts $\xi_1, \xi_2 \in \Omega$*—chosen once and for all from (s_0, m, a_1, a_2),* independent of K_1, K_2, and q *(with a* single *odd shift $\xi \in \Omega$, chosen solely from (s_0, m, a_1), sufficing whenever $a_1 = a_2$ and $K_1 = K_2$)—such that for* every $K_1, K_2 \in \mathbb{Z}$ *and* every *modulus $q \geq 3$, setting $\tilde{Q}_h = Q(a_h, K_h \bmod q)$, at least one of the following three failure regimes occurs:*

1. (Small-q super-Fibonacci regime, $3 \leq q < 32$.) *For every $k_0 \in \mathbb{N}$, the window $[k_0, k_0 + 11)$ of* 11 *consecutive levels on head* 1 *contains at least one level whose reduced blind apex error or reduced blind band-*$0, 1, 2$ *error strictly exceeds $q/8$ (realized as walk valley-band hop errors whenever $1 \leq k_0$ and $k_0 + 10 \leq m$).*

2. (Large-$\tilde{Q}$ blind fine-overflow regime, $q \geq 32$ and $q < 4\tilde{Q}_h + 4$ for some $h \in \{1, 2\}$.) *On head h, there is a served residue* $\text{rem} < a_h$ *such that* every *level $k < m$ with $(m-k) \bmod (a_h+1) = \text{rem}$ has reduced blind apex error $\geq q/16$, or reduced blind band-*0 *or band-*1 *error $\geq q/8$.*

3. (Small-$\tilde{Q}$ coarse V-straddle regime, $q \geq 32$ and $4\tilde{Q}_1 + 4 \leq q$, $4\tilde{Q}_2 + 4 \leq q$.)

   - *If $\tilde{Q}_1 \leq \tilde{Q}_2$, then at odd shift ξ_1, at least $\frac{1}{16}|K_{\text{coarse}}|$ coarse levels $k \in K_{\text{coarse}}$ of head* 1 *simultaneously satisfy, against $\kappa = \texttt{cancellerBelow}(m, a_2, k)$ of head* 2*,*

$$\begin{aligned} \texttt{classStraddleCount}(s_0, m, k, \kappa, \text{true}, \xi_1) &\geq \frac{2^{k+2} R_k}{2^{a_2+7}}, \\ (2A_k^{(1)} + 2(c_{\text{f}}^{(2)}(\kappa) - c_{\text{b}}^{(2)}(\kappa))) \bmod q &\geq \frac{q}{2}. \end{aligned}$$

   - *Symmetrically, if $\tilde{Q}_2 < \tilde{Q}_1$, then at odd shift ξ_2, at least $\frac{1}{16}|K_{\text{coarse}}|$ coarse levels $k \in K_{\text{coarse}}$ of head* 2 *satisfy the analogous bounds with a_1 and $(B(a_2, K_2), B(a_1, K_1))$.*

In both subcases of Regime 3, combining the straddle count and residue lower bounds of Theorem 5.5(3) with the per-V straddle inequality (6) of Theorem 5.2(2) shows that at every such coarse level $k \in K_{\text{coarse}}$, the straddling apex hops accumulate total reduced mod-q hop error at least $\Omega(2^{-a} q \cdot 2^k R_k)$—a constant fraction $2^{-O(a)}$ of the entire class capacity $q \cdot 2^{k+1} R_k$.

Outline of the formal proof. We highlight the four formal components in `DNPConjectureApp.lean` (which also establishes prefix lockstep $\alpha_\kappa(\varepsilon) - \alpha_k(\varepsilon)$ independent of $(\varepsilon_1, \ldots, \varepsilon_{k-1})$, `apexOffset_diff_indep_prefix`):

1. *Exact odd-shift straddle sum (*`sum_classStraddleCount_odd`*).* The $2^{\kappa+1}$ level-κ V-spans $[\alpha_V, \alpha_V + 2\Pi_\kappa)$ in $[0, P)$ are pairwise disjoint (`periodVApices_spans_disjoint`, `card_periodVApices`) and have even start times when $m < s_0$ (`even_of_mem_periodVApices`). For each coarse apex c of level k, as ξ ranges over the $P/2$ odd shifts $\texttt{oddRange}(P)$, $(c + \xi) \bmod P$ (and likewise $(c + P - \xi) \bmod P$) bijects onto the $P/2$ elements of $[0, P)$ of parity opposite to c. Summing over α_V and applying Theorem 5.2(3) on each $[\alpha_V, \alpha_V + 2\Pi_\kappa)$ yields the exact identity

$$\sum_{\xi \in \texttt{oddRange}(P)} \texttt{classStraddleCount}(s_0, m, k, \kappa, \text{dir}, \xi) = 2^{k+1} \cdot 2^{\kappa+1} \cdot (\Pi_\kappa/2)^2.$$

2. *Simultaneous reverse-Markov selection (*`reverse_markov_simultaneous` *and* `exists_oddShift_coarse_straddle`*).* For every $k \in K_{\text{coarse}}$ and $\kappa = \text{cancellerBelow}(m, a, k)$

(`cancellerBelow`, `cancellerBelow_props`), we have $\Pi_\kappa \le R_k \le 2^{a+2}\Pi_\kappa$ and $m-\kappa+1 \ge \frac{1}{2}(m+1)$, so letting $X(\xi,k)$ denote `classStraddleCount` at $(s_0,m,k,\kappa,\mathrm{dir},\xi)$ gives $X(\xi,k) \le M_k := 2^{k+1}\Pi_\kappa$ (because at most one V-span contains any shifted center, `totalStraddleAtCenter_le`), while its mean over $\Omega = \texttt{oddRange}(P)$ ($|\Omega| = 2(m+1)2^{s_0}$) satisfies $2^{\kappa+1}(\Pi_\kappa/2)^2/|\Omega| = \frac{m-\kappa+1}{4(m+1)}\Pi_\kappa \ge \frac{1}{8}\Pi_\kappa$, giving $\mathbb{E}_{\xi\in\Omega}[X(\xi,k)] \ge \frac{1}{8}M_k$. Applying reverse Markov per level k (`reverse_markov_single`) gives $\Pr_{\xi\in\Omega}[X(\xi,k) \ge \frac{1}{16}M_k] \ge \frac{1}{16}$, and double-counting over K_{coarse} (`reverse_markov_simultaneous`) exhibits a *single* odd shift $\xi \in \Omega$ at which at least $\frac{1}{16}|K_{\mathrm{coarse}}|$ coarse levels satisfy $X(\xi,k) \ge \frac{1}{16}M_k \ge 2^{k+2}R_k/2^{a+7}$.

3. *Mod-q straddle residue without wrap-around (*`block_straddle_residue_ge_half_q`*).* Suppose $q \ge 32$, $\tilde{Q}_1 \le \tilde{Q}_2$, and $4\tilde{Q}_2 + 4 \le q$. Let $\tilde{K}_h = K_h \bmod q \in [0,q)$. At any level k of $B(a_1,\tilde{K}_1)$, $A_k^{(1)} \in \{0,\tilde{K}_1\}$, while $2\tilde{K}_1 + 2 \le \tilde{Q}_1 \le \tilde{Q}_2$ (since $a_1 \ge 1$). At canceller κ of $B(a_2,\tilde{K}_2)$, $c_{\mathrm{f}}^{(2)}(\kappa) - c_{\mathrm{b}}^{(2)}(\kappa) = -\tilde{Q}_2 - 1$. Therefore the integer $Z := 2A_k^{(1)} + 2(-\tilde{Q}_2 - 1)$ satisfies

$$-\frac{q}{2} \;\le\; -2\tilde{Q}_2 - 2 \;\le\; Z \;\le\; 2\tilde{K}_1 - 2\tilde{Q}_2 - 2 \;\le\; -4 \;<\; 0.$$

   Because $Z \in [-q/2,-1]$, its Euclidean remainder $Z \bmod q = Z + q$ lies in $[q/2, q-4]$, hence $\ge q/2$.

4. *Fine-scale doubling overflow (*`block_overflow_band_witness`*).* Suppose $q \ge 32$ and $4\tilde{Q}_h + 4 > q$. Set $\tilde{K} = K_h \bmod q \in [0,q)$. If $16\tilde{K} \ge q$, every served level (rem $= 0 < a_h$) has $a(k) = \tilde{K} \ge q/16$. If $16\tilde{K} < q$, consider the doubling sequence $T_r = (2^r - 1)(2\tilde{K} + 2)$ for $0 \le r \le a_h$: we have $T_0 = 0$, $T_{r+1} = 2T_r + (2\tilde{K} + 2)$, and $4T_{a_h} + 4 = 4\tilde{Q}_h + 4 > q$. Since $2\tilde{K} + 2 \le q/8 + 2 < q/4$ for $q \ge 32$, the first index $r \le a_h$ where $\tilde{K} + T_r \ge q/8$ cannot jump past $7q/8$, landing strictly inside $[q/8, 7q/8] \subseteq [q/8, q-1]$. When $r < a_h$, $\tilde{K} + T_r = E_k \bmod q = b_0(k)$ on residue rem $= r$; when $r = a_h$, $\tilde{K} + T_{a_h} = (E_k - (A_{k+1} + E_{k+1} + 2)) \bmod q = b_1(k)$ across the canceller $k+1$ on residue rem $= 0$.

Combining Regimes 1 (from Theorem 5.3(2) with $s = 11 > 2\lfloor \log_2 31 \rfloor + 2 = 10$), 2, and 3 proves `theoremA_three_regimes` and `theoremA_symmetric_blockPair`. □

Remark 5.6 (Scope relative to Conjecture A″)**.** Theorem 5.5 gives three-regime failure witnesses for block-label pairs $(B(a_1,K_1), B(a_2,K_2))$ of bounded block length across all $K_1, K_2 \in \mathbb{Z}$ and all $q \ge 3$ with *zero logarithmic loss in* q, at two fixed odd shifts (one shift for symmetric pairs $a_1 = a_2, K_1 = K_2$). Deriving Conjecture A″ [7, Section 5.6, Conjecture 5.24] for block pairs—and ultimately for all low-cost labels on the r-fold cover—requires four further steps:

(a) *One common odd shift for asymmetric pairs.* Conjecture A″ fixes one odd shift ξ defining the pair $x_2 = x_1(\cdot + \xi)$ before quantifying over labels and q. In Regime 3 of Theorem 5.5, ξ_1 is used when $\tilde{Q}_1 \le \tilde{Q}_2$ and ξ_2 when $\tilde{Q}_2 < \tilde{Q}_1$, which depends on (K_1, K_2, q); applying first-moment reverse Markov separately to dir = true and dir = false gives two subsets of Ω of density at least $\frac{1}{16}$ that need not intersect without a joint second-moment (carry-decorrelation) bound across `periodVApices`.

(b) *Half-blindness at the fixed shift.* Regimes 1 and 2 bound *blind* single-head errors (`blindApexErr` and `blindBandErr`, where $F = 0$), which coincide with actual two-head hop errors only on hops whose window contains no turn of the other head. In [7, Proposition 5.17 and Corollary 5.18], blindness of fine classes is proved on average over shifts rather than at the specific shift chosen for Regime 3.

(c) *From hop-error witnesses to doubly θ_0-served level counts.* Conjecture A″ bounds the number of doubly θ_0-served levels by $C \log_2 q + C' + \eta(m+1)$ ($\eta < 1$). Passing from Theorem 5.5's witnesses to that level-count bound requires combining (b) on fine levels with the explicit class-fraction weights of the apex and valley-band hop families in Regimes 1–3 to determine $\theta_0(a_0)$ and $\eta < 1$.

(d) *Unbounded block lengths and general low-cost labels.* Beyond bounded block lengths $a_1, a_2 \le a_0$, when $\log_2 q \ll m \ll q$ and $a \asymp \frac{1}{2}\log_2 q$, the single-canceller straddle density $2^{-a} \asymp q^{-1/2}$ drops below constant θ_0; and full Conjecture A″ covers *every* label of cost at most $n_{\mathrm{h}}/8$ on every r-fold cover (not only turn-supported or level-uniform block potentials, and subject to period closure modulo rP).

Remark 5.7 (Finite numerical illustration)**.** The standalone script `tools/reproduce_numerics.py` evaluates exact stroke counts, odd-center straddle sums, super-Fibonacci band unwrapping, exhaustive odd-shift straddle sums (e.g., on $s_0 = 7, m = 6$), and all three regimes of Theorem 5.5 on finite parameters (e.g., $s_0 = 24, m = 23, a_1 = a_2 = 1$, where $4(\lfloor \log_2 24 \rfloor + 1 + 1) = 24 = m + 1$ and $K_{\mathrm{coarse}} = \{1, \dots, 6\}$), confirming that the explicit odd-shift reverse-Markov witness and mod-q straddle residues match the formal bounds across all tested (K_1, K_2, q).

6 What a linear-time transposer would need

Our aim across this series is to decide (T), whether by constructing a linear-time transposer or by proving that none exists. Impossibility theorems and candidate constructions inform one another directly: every conditional or restricted half-rate bound and label-cut barrier rules out an entire mechanism on which a linear-time machine cannot rely, narrowing the design space to the exact structural regimes where impossibility is not yet certified. This section collects what the results of Sections 3 to 5 require of any linear-time transposer (oblivious unless stated otherwise) and identifies the specific mechanisms that fail. It introduces no new numbered theorems: every claim is one of the formalized results above, a cited result of an earlier paper in the series, or explicitly marked as a heuristic.

6.1 The depth–coding interface

For an oblivious transposer M with k work tapes, value width β, and running time $T \le C(L+1)$ at input length $L = d^2 b(d)$ ($b(d) = \max(1, \lceil \log_2 d \rceil)$), Theorem D of the fourth paper [8, Section 4.2, Theorem 4.10] (proved in the text of [8]) presents a dichotomy for every $\varepsilon \in (0, 1]$ and $D \ge 0$:

- *Either* the dependency DAG H_M admits no genie of cost at most εL and depth at most D (defeating cheap depth reduction);
- *or* the revealed word of such a genie forms an index code of at most εL bits for the bounded-depth cone side information N_I on $[0, L)$, and its largest fibre is a binary storage code $C \subseteq \{0,1\}^L$ of rate at least $1 - \varepsilon$ on N_I [8, Remark 4.9(b)].

By the fifth paper's paired-distance analysis [12, Theorem 3.3 and Corollaries 3.7 and 3.8], for every fixed $\delta > 0$ and odd radius r, deleting a set Z of at most $\delta L + O(L/\log d)$ coordinates (high-reuse positions together with short paired-incidence paths, measured by the short-pair count $b_E(r)$ in the bipartite incidence graph of N_I) leaves a bounded-degree relation $N_I - Z$ with $b_E(r) = 0$; at $r = 3$, $N_I - Z$ is loopless and TT-free [10, Lemma 2.1] (`HalfRate.ttFree_iff_shortPaired_eq_empty`). Moreover, by paid coordinate deletion ([10, Corollary 6.9], Lean-verified as `HalfRate.card_sq_`

`le_of_paidDeletion`), if $N_I - Z$ satisfies the polymatroid half-rate bound (1), then $|C|^2 \le 2^{L+|Z|}$, bounding the rate of C by $\frac{1}{2} + |Z|/(2L) \le \frac{1}{2} + \frac{\delta}{2} + o(1)$ and ruling out rate $1-\varepsilon$ whenever $\varepsilon < \frac{1}{2} - \frac{\delta}{2}$.

6.2 Necessary conditions on the cone codes

Suppose an oblivious linear-time transposer admits a bounded-depth genie of cost εL ($\varepsilon < \frac{1}{2}$), yielding a binary storage code C of rate at least $1-\varepsilon$ on N_I, and let $N' = N_I - Z$ be the loopless TT-free sub-relation obtained after paid deletion of $|Z| \le \delta L + o(L)$ coordinates. In addition to the conditions established in [8, 12, 10, 7], our coding theorems impose two sharp structural requirements on N':

(N1) *Positive density of unreciprocated full-in-degree vertices ($\neg\mathcal{S}$).* For *every* subset W of coordinates of N' on which the induced relation $N'[W]$ belongs to $\mathcal{S}$, we must have

$$|W| \;\le\; (2\varepsilon + \delta)L + o(L).$$

Indeed, applying Theorem 3.6(2) (`rateBound_classS`) to $N'[W]$ and paid deletion to the remaining $L - |W|$ coordinates of N_I gives $|C|^2 \le 2^{L+(L-|W|)} = 2^{2L-|W|}$, whereas $|C| \ge 2^{(1-\varepsilon)L}$ forces $2(1-\varepsilon)L \le 2L - |W|$, i.e., $|W| \le 2\varepsilon L$ (and at most $(2\varepsilon+\delta)L + o(L)$ inside N'). Likewise, by Corollary 3.7, on any $\mathcal{S}$ sub-relation no vertex cut S can create an isolated-parent surplus $|I_{N'[W]}(S)| - |S|$ without correspondingly shrinking code capacity.

Design consequence: When N' has maximum in-degree 3, a coding gain above rate $\frac{1}{2}$ (whether linear or nonlinear, over any alphabet) cannot come from subcubic undirected exchange, pure one-way directed cycles of in-degree-2 vertices, chordal one-way shortcuts across digon cycles, or even stride-3 hub cycles such as those of G_{21} (Proposition 3.8) whose hubs have in-degree 2. Rather, a linear fraction of coordinates in strongly connected components of N' must have *in-degree* 3 *together with at least one unreciprocated outgoing arc* ($v \in N'(u)$ with $u \notin N'(v)$), or else have in-degree at least 4.

(N2) *Distance-2 heavy clusters for degree-4 scalar linear codes.* Suppose the machine's data path is linear over a field F and a fibre code on a triangle-free symmetric support $G = (V, E)$ of maximum degree 4 is realized as $\ker M$ for a unit-diagonal check matrix $M \in F^{V \times V}$. By Theorem 4.1 and Corollary 4.2 (`finrank_ker_le_matching_rhoMatch`),

$$\dim_F \ker M \;\le\; \nu(G) + |H_M| - \rho_{\mathrm{m}}(G, H_M) \;\le\; \frac{|V|}{2} + |H_M| - \rho_{\mathrm{m}}(G, H_M),$$

and the same bound holds with $H_{M^\intercal}$ in place of H_M. To achieve $\dim_F \ker M \ge (1-\varepsilon)|V|$, the code must satisfy

$$|H_M| - \rho_{\mathrm{m}}(G, H_M) \;\ge\; \Big(\frac{1}{2} - \varepsilon\Big)|V|, \qquad |H_{M^\intercal}| - \rho_{\mathrm{m}}(G, H_{M^\intercal}) \;\ge\; \Big(\frac{1}{2} - \varepsilon\Big)|V|.$$

In [7], where $\rho_3(G, H_M)$ stood in place of $\rho_{\mathrm{m}}(G, H_M)$, adjacent pairs of degree-4 rows remained a plausible source of linear excess (as on the bipartite family G_{8k} of Remark 4.3, where $|H| - \rho_3(G_{8k}, H) = k = \frac{1}{8}|V|$). Corollary 4.2 closes that loophole completely: isolated heavy edges ($u \sim v$ with no shared neighbor, since G is triangle-free) contribute full weight 2 to $\rho_{\mathrm{m}}(G, H_M)$ and yield *zero* rate gain above $\frac{1}{2}$. Any degree-4 scalar linear gain must concentrate a linear fraction of heavy vertices at pairwise distance *exactly* 2 in G, sharing common neighbors.

6.3 Necessary conditions on head schedules and cut labels

By Theorems F and M of [7], any oblivious linear-time transposer that defeats cheap depth reduction must employ unbounded owner-load on at least two work heads, or on one work head coupled with bounded-load heads across unboundedly many separated scales; the 2-head deep-nesting pair $\mathrm{DNP}(s_0, m)$ is the minimal prototype of the two-head regime. Our results in Section 5 sharpen both sides of the $\mathrm{DNP}(s_0, m)$ frontier:

- *Why simple turn-label cuts fail to prove* TCP(2) *on* $\mathrm{DNP}(s_0, m)$. An impossibility proof via TCP(2) must exhibit, for every $\varepsilon > 0$, a bounded modulus $q = O_\varepsilon(1)$ and a low-cost label $\Psi : [0, T) \to \mathbb{Z}_q$. Theorem 5.3 proves that no level-uniform turn potential can keep blind apex and band-$0, 1, 2$ errors at most $q/8$ across more than $2\lfloor \log_2 q \rfloor + 2$ consecutive levels without periodic negative resets; Proposition 5.4(2)–(3) proves that periodic $(a+1)$-block resets carry strictly positive mean path drift $+\frac{1}{2}(a(K+1)+1)$ across random nested branches while pure binary clocks modulo Mersenne numbers $q = 2^B - 1$ suffer $q - 1$ straddle error whenever $B \mid (m - \kappa + 2)$; and Theorem 5.5 exhibits three-regime blind-error and coarse-straddle witnesses on $\Omega(m)$ levels at two fixed odd shifts ξ_1, ξ_2 (a single odd shift for symmetric pairs) for *every* bounded-range block pair $(B(a_1, K_1), B(a_2, K_2))$ across *all* $K_1, K_2 \in \mathbb{Z}$ and *all* $q \geq 3$. Thus any cut refuting depth-robustness of $\mathrm{DNP}(s_0, m)$ at misaligned odd shifts where fine bands are half-blind must exploit mechanisms outside Theorem 5.5 (Remark 5.6), such as block lengths a growing with $\log_2 q$, non-level-uniform jumps varying across sibling V's at the same scale, or cuts not supported at turns.

- *What a depth-robust* 2*-head schedule must do (heuristic reading).* Conversely, if a 2-head oblivious transposer seeks to evade cheap depth cuts through deep nesting, aligned shifts ($\xi = 1$) fail because both heads share a single doubled block potential [7, Proposition 5.14], whereas misaligned odd shifts create the simultaneous multi-scale V-straddle interference quantified by Theorem 5.2 and Theorem 5.5: coarse apex hops of scale R_k straddle canceller valleys of scale $\Pi_\kappa \asymp R_k$ on the other head with exact parity-invariant weight $\Pi_\kappa^2/4$ per V (Theorem 5.2(3)). Even if such a schedule achieves depth-robustness on its deep core, Theorem F of [7] restricts that core to $o(L)$ heavily visited scratch cells unless the machine also overcomes the Scratch Question ([7, Section 10, Problem 10.5])—or, for adaptive machines, escapes the complete source-cover and return-profile barriers of [14].

7 Open problems

7.1 Updated map of half-rate and deep-nesting bounds

For oblivious machines, Theorem D [8, Section 4.2, Theorem 4.10] reduces ¬(T) to depth reduction (TCP(k)) together with a sufficiently strong coding deficit on the actual cone relations. The proposed abstract half bound [12, Question 9.1] is false [11]; its native form remains open. The table below updates the status of exact radius-3 (loopless TT-free) half-rate bounds from [10, Section 9.1] and [7, Section 8]:

regime	proved	open nearby
directed in-degree ≤ 3, full reciprocal closure at in-degree 3 ($\mathcal{S}$)	all codes, all alphabets: piece decomposition and $\lvert C\rvert^2 \leq \lvert\alpha\rvert^{\lvert V\rvert}$ (**Theorem S***, Theorem 3.6)	general TT-free in-degree ≤ 3 (Problem 7.1; Conjecture P of [10])
chordless digon cycles with ≤ 1 outside parent per vertex	piece certificate exists for some $m \in \mathbb{N}$ iff not a stride-3 hub cycle (**Theorem DC**, Theorem 3.4); X_{15} core and $GP(5,2)$ ruled out for all $m \in \mathbb{N}$ (Corollary 3.3 and Proposition 3.5)	chorded digon cycles with one-way chords (Conjecture K of [10]); $O(1)$ record bound on $\mathcal{S}$ (Problem 7.2)
triangle-free degree ≤ 4, scalar linear ($\ker M$, unit diagonal)	$\dim_F \ker M \leq \lfloor \lvert V\rvert/2 \rfloor$ under (S1) alone (heavy matchings allowed; **Theorem SH***, Theorem 4.1), plus ρ_{m} barrier (**Corollary M***, Corollary 4.2)	degree 4 with growing odd girth; the general finite half bound is false (Problem 7.3)
DNP(s_0, m), turn-supported mod-q labels	blind-error run $\leq 2\lfloor \log_2 q \rfloor + 2$ (**Theorem LU**, Theorem 5.3); three-regime failure witnesses for bounded-range block pairs at two odd shifts, one for symmetric pairs (**Theorem A**, Theorem 5.5)	common asymmetric shift, fixed-shift blindness, served-level conversion, $a \asymp \frac{1}{2}\log_2 q$, and general TCP(2) (Problem 7.4)

7.2 Open problems

1. *Decide (T).* Construct a deterministic multitape Turing machine that transposes $d \times d$ matrices of $b(d)$-bit entries ($b(d) = \max(1, \lceil \log_2 d \rceil)$) in time $O(d^2 b(d))$, or prove that none exists (checked against `Transpose.LinearTimeTransposable` in [9]). On adaptive machines, extending the complete source-cover and return-profile barriers of [14] beyond restricted return geometries remains the central non-oblivious barrier question.

2. *Conjecture P beyond $\mathcal{S}$, and Conjecture K.* Theorem 3.6 resolves the intermediate target named in item 2 of [10, Section 9.2] (the all-code version of the fourth paper's Theorem S on its whole class). Conjecture P ([10, Conjecture 9.1]) remains open; the gap to the full directed degree-3 case lies in vertices of in-degree 3 that send unreciprocated outgoing arcs:

   Problem 7.1 (Directed in-degree three without reciprocal closure)**.** *Does every finite loopless TT-free recovery relation with $\lvert N(v)\rvert \leq 3$ for all $v \in V$ admit a certified piece decomposition (`HalfRate.Decomp`, Conjecture P of [10, Section 9.2]), or at least satisfy the polymatroid half-rate bound $2h(V) \leq \sum_{v\in V} h(\{v\})$? As a concrete special case (Conjecture K of [10, Section 9.2]), does every loopless TT-free chorded digon cycle with at most one incoming and at most one outgoing one-way chord per vertex admit a global elimination certificate?*

   By [10, Proposition 8.1] (`HalfRate.dn3_rate_eventually_le`), an affirmative answer to Problem 7.1 implies an affirmative answer to the paired-distance coding question [12, Question 9.1] for all recovery relations of in-degree at most 3, with no restriction on out-degree. On $X_{15} \notin \mathcal{S}$, vertex 6 has in-degree 3 and sends a pure one-way arc $6 \to 3$, obstructing its minimum core $\{0, \ldots, 5\}$ across all $m \in \mathbb{N}$ (Corollary 3.3), yet X_{15} still admits a 3-record global certificate [10, Remark 7.3].

3. *Record complexity of piece decompositions in $\mathcal{S}$.* In Proposition 3.5 we showed that global elimination certificates exhibit a strict record hierarchy ($m = \infty$ on GP$(5, 2)$, $m = 4$ on GP$(7, 2)$, and $m = 3$ on GP$(9, 2)$). For piece decompositions in $\mathcal{S}$, pure stride-s cores ($s \geq 4$) use $m \leq 3$ records (Proposition 3.2(2)), whereas in Theorem 3.4 and Theorem 3.6 our certificates on non-stride-3 chordless digon L-cycles ($L \geq 5$), minimal chordal arcs of span $d \geq 3$ (Subcase 4a), and hub-augmented stride-3 g-cycles (Subcase 4c) use up to $L + 1$, $d + 2$, and $g + 2$ records, respectively.

   Problem 7.2 (Bounded-record decompositions for $\mathcal{S}$)**.** *Is there an absolute constant m_0 (for instance $m_0 = 3$ or $m_0 = 4$) such that every finite relation $N \in \mathcal{S}$ admits a certified piece decomposition in which every piece certificate uses at most m_0 records?*

4. *Triangle-free storage codes at degree* 4 *with distance-*2 *heavy clusters.* Theorem 4.1 and Corollary 4.2 show that for scalar linear codes $\ker M$ on triangle-free graphs of maximum degree 4, heavy vertices need not be non-adjacent: only shared neighbors between distinct heavy vertices ($\mathrm{dist}_G(u, v) = 2$, violating (S1)) can break the minimal-counterexample run.

   Problem 7.3 (Distance-2 heavy clusters and all-code degree-4 bounds)**.** *(a) Over an arbitrary field F, does $\dim_F \ker M \leq \lfloor |V|/2 \rfloor$ hold for every unit-diagonal matrix M supported on a finite triangle-free graph G of maximum degree* 4*, when H_M contains vertices at distance* 2 *in G—or at least up to $o(|V|)$ along locally bipartite families of growing odd girth? (b) Does the polymatroid half-rate bound $2h(V) \leq \sum_{v \in V} h(\{v\})$ hold on every triangle-free undirected graph of maximum degree* 4*?*

   Update (November 2026). The finite assertion in (a) and the assertion in (b) are false by [10, Example 7.8]. That triangle-free 4-regular graph has 13 vertices and an $\mathbb{F}_4$-linear storage code C of dimension 7. Its linear recovery equations give a matrix $M = I - A$ with unit diagonal, supported on the graph, and $C \subseteq \ker M$, so $\dim \ker M \geq 7 > \lfloor 13/2 \rfloor$. By Theorem 4.1, its heavy set must contain two vertices with a common neighbor; triangle-freeness makes their distance exactly two, as required in (a). Taking the entropy of the uniform codeword also refutes the polymatroid bound in (b). The storage-code counterexample is Lean-verified in the seventh paper; the matrix and entropy interpretations here are written consequences. Its odd girth is five, so the growing-odd-girth alternative in (a) remains open. At degree 5, scalar binary codes already reach rate 5/8 on the Clebsch graph (Lean-verified in [8, Corollary 6.11]), whereas at degree 4 tree-local arguments cannot handle arbitrary clusters of heavy vertices [8, Proposition 6.28].

5. *Closing the block-pair gaps to Conjecture A″, and* TCP(2) *on the deep-nesting pair.* Theorem 5.5 gives three-regime failure witnesses for block-label pairs $(B(a_1, K_1), B(a_2, K_2))$ of bounded block length on DNP(s_0, m) across all $K_1, K_2 \in \mathbb{Z}$ and $q \geq 3$ at two fixed odd shifts (one shift for symmetric pairs). On the r-fold periodic cover of DNP(s_0, m) (containing $n_{\mathrm{h}} = 2rP = 8r(m + 1)2^{s_0}$ long hops across the two heads), the four steps of Remark 5.6 remain to establish Conjecture A″ [7, Section 5.6, Conjecture 5.24] and decide TCP(2):

   Problem 7.4 (From Theorem A to Conjecture A″ and walk-DAG depth of DNP)**.** *(a) Prove the existence of a* single *odd shift $\xi \in \Omega$ that works simultaneously for both coarse straddle directions (*$\mathrm{dir} = \mathrm{true}$ *and* $\mathrm{dir} = \mathrm{false}$*) and satisfies half-blindness on fine sub-bands, and deduce from Theorem 5.5 an explicit upper bound $C \log_2 q + C' + \eta(m + 1)$ ($\eta < 1$) on doubly θ_0-served levels for all bounded-range block pairs. (b) Extend the bound when q is exponential in m*

and $\max(a_1, a_2) \asymp \frac{1}{2}\log_2 q$*, and from block pairs to* every *mod-q label of cost at most $n_{\mathrm{h}}/8$ on the r-fold cover (full Conjecture A″). (c) Are the walk DAGs of* $\mathrm{DNP}(s_0, m)$ *depth-robust as $m \to \infty$ at some sequence of odd shifts, refuting* $\mathrm{TCP}(2)$ *(and hence* 4*-stack depth reduction* $\mathrm{SR}(4)$*; compare* $\mathrm{SR}(3)$ *and the Scratch Question in [7, Section 10, Problems 10.2 and 10.5])?*

A Index of Lean declarations

This index lists every declaration of this paper's library cited in the text, relative to the namespace `DirectedReciprocity`. The declarations named in status lines are grouped by the module named there; those named only in the text are listed at the end. All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). Declarations of earlier libraries of the series that are named in the text, in the namespaces `Transpose`, `MotionDepthCodes`, `DirectedRecovery`, `HalfRate`, `BoundedNesting` and `SimpleGraph` (`lean-machine`), are defined and verified in those libraries [9, 8, 12, 10, 7]; all except the standalone `Transpose` package of [9] are also re-checked directly in `lean/PaperAxioms.lean`.

`DirectedReciprocity.ClassS` `ClassSOn`, `IsClassS`, `barrier_ineq_classS`, `card_sq_le_classS`, `classSOn_decomp`, `classS_decomp`, `exists_inClosed_scc`, `exists_piece_of_scc_classS`, `isolatedOutDir`, `liftPieceCertInClosed`, `log_card_le_barrier_classS`, `rateBound_classS`.

`DirectedReciprocity.ClassSExamples` `g21Rel`, `g21_hex_no_pieceCert`, `g21_isClassS`, `g21_min_indeg_and_acyclic_W`, `g21_no_any_triangle`, `g21_shortest_digon_cycles`, `g21_stronglyConnected`, `rateBound_g21`, `x15_hexagon_no_pieceCert`.

`DirectedReciprocity.DNPConjectureApp` `apexOffset_diff_indep_prefix`, `blockA`, `blockE`, `blockQ`, `block_blind_errors_mod_K`, `block_canceller_mass`, `block_overflow_band_witness`, `block_served_mass`, `block_straddle_residue_ge_half_q`, `cancellerBelow_props`, `card_periodVApices`, `classStraddleCount`, `even_of_mem_periodVApices`, `exists_oddShift_coarse_straddle`, `periodVApices_spans_disjoint`, `propOC_back_bounds`, `propOC_fwd_bounds`, `propOC_mersenne_straddle`, `propOC_sum_all_paths`, `reverse_markov_simultaneous`, `sum_classStraddleCount_odd`, `theoremA_symmetric_blockPair`, `theoremA_three_regimes`.

`DirectedReciprocity.DNPFibonacci` `blindApexErr`, `blindBandErr`, `blind_error_le_div_eight`, `log_bound_of_fib_le`, `strokePhiRec_apex_hop`, `strokePhiRec_valley_band_hop`, `superFib_lower_bound`, `superFib_step`, `theoremLU_consecutive`, `unwrap_blindBand_step`.

`DirectedReciprocity.DNPStraddle` `HasLevelUniformVDecomp`, `mem_straddleSet_sub_iff`, `straddleSet`, `straddle_otherHeadInc_sum`, `strokePhiRec_hasLevelUniformVDecomp`, `sum_straddleSet_card_parity`, `theoremVStar_pair`, `theoremVStar_sum`.

`DirectedReciprocity.DNPStrokes` `card_stroke_stopDepth`, `card_stroke_stopDepth_branchWord`, `card_stroke_stopDepth_branchWord_parity`, `card_stroke_stopDepth_parity`, `decompDepth_valid`, `decomp_reconstruct`, `dnpPeriod`, `evalPhiRec`, `evalPhiRec_eq_sum_turnJumps`, `reconstruct_decomp`, `scaleR`, `strokeLen`, `strokePhiRec`, `strokePhiRec_back`, `strokePhiRec_fwd`.

`DirectedReciprocity.DigonCycle` `IsStride3Cycle`, `digonCycle4_pieceCert`, `digonCycleGe5_pieceCert`, `digonCycle_pieceCert_iff`, `digonCycle_pieceCert_of_not_stride3`, `exists_non_stride3_pos`, `not_pieceCert_of_isStride3Cycle`.

DirectedReciprocity.PetersenHierarchy generalizedPetersen_hierarchy, gp52_eq_petersenGraph, gp52_no_elimCert, gp72_elimCert4, gp72_no_elimCert_le3, gp92_elimCert3, gpGraph, nonempty_elimCert_iff_pieceCert_univ, petersen_no_pieceCert_univ.

DirectedReciprocity.PositionalNest PieceCert.digon_ne, PieceCert.mem_supply_of_idxOf_lt, disjoint_nested_iff, relCharged_iff_nested, relCharged_of_nested.

DirectedReciprocity.StrideCycles stride3_no_pieceCert, strideCore, strideCore3_no_pieceCert, strideCore_pieceCert_iff, strideEven_pieceCert_two, strideGe4_pieceCert, strideRel.

DirectedReciprocity.TheoremSHStar F3_star, MinCE.false_SHStar, NoCommonNbhd, SHStarOn, exists_rhoMatch, finrank_ker_add_card_le_matching_SHStar, finrank_ker_le_barrier_SHStar, finrank_ker_le_matching_rhoMatch, finrank_ker_le_matching_rhoMatch_transpose, finrank_subCode_le_half_SHStar, odd_state_star, rho3_le_rhoMatch, rhoMatch, theoremSHStar, theoremSHStar_graph, theoremSHStar_transpose.

Also named in the text SHStarOn.mono, ValidStrokeCode, augmentedStride3_pieceCert, blockPathSum, cancellerBelow, card_Ico_even_len_mod_two, cycRev, dcCol, dcPerm, dcSupply, decompDepth, digonGraph, digonGraph_cliqueFree3, elimCert_adj_ne, isCanceller, minChordArc_pieceCert, not_mem_rev_empty_of_lt_idxOf, oddRange, otherHeadInc, pathSumBack, pathSumFwd, periodTurnJump, periodVApices, reconstructDepth, relCharged_step_nested, relCharged_step_none, relCharged_step_some, resRel, reverse_markov_single, strokeLen_succ, strokeMass, strokePhiRec_eq_sum_turnJumps, strokeTurnJump, sum_fin_two_pow, sum_min_sub_two_mul, totalStraddleAtCenter_le, two_pow_add_one_le_fib.

B Building and verifying the formalization

The Lean 4 formalization lives in `lean/` next to this paper. It uses Lean and Mathlib at version `v4.34.1` (`lean/lean-toolchain`, `lean/lakefile.toml`, `lean/lake-manifest.json`) and builds directly on the Lean libraries of the seventh paper (`halfRate`, [10]) and the eighth paper (`boundedNesting`, [7]) via relative paths; through them it also imports the libraries of the fourth, fifth, second, and first papers [8, 12, 13, 9] together with the repository's shared library `lean-machine`. Keep the repository's sibling directory layout intact so that these relative package paths resolve.

Build. On a fresh clone of the repository, first fetch the second paper's pinned external dependency as documented in its README (`python3 verification/fetch_openai_metric.py`, run inside `../Local-Coding-Barriers-and-the-Limits-of-Tape-Geometry-for-Matrix-Transposition/`). Then, with `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads precompiled Mathlib `.olean` caches; the second compiles all eleven modules of `lean/DirectedReciprocity/` and the root module `lean/DirectedReciprocity.lean` at Lean's default `maxHeartbeats` setting (200,000), with zero errors and zero warnings.

Axioms and declaration index. The script `tools/lean_index.py` scans `paper/sections/*.tex` and regenerates both the declaration index of Appendix A (`paper/sections/leanindex.tex`) and `lean/PaperAxioms.lean` (pass `-index-only` to write only the index). Running in `lean/`:

```
lake env lean PaperAxioms.lean
```

executes `#print axioms` on every declaration of `DirectedReciprocity`—and every declaration of earlier libraries—cited in this paper. Every declaration depends at most on the three standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`. No file of this library uses `sorry`, `admit`, `native_decide`, `unsafe`, or any custom `axiom`; finite graph and partition checks in Section 3 are verified inside Lean's kernel via `decide`, `decide +kernel`, and `decide +revert`.

Standalone numerical cross-checks. The portable Python 3 script `tools/reproduce_numerics.py` uses only the Python standard library and independently reproduces the finite numerical illustrations of Remarks 4.3 and 5.7:

```
python3 tools/reproduce_numerics.py
```

Specifically, it constructs the bipartite degree-4 family G_{8k} ($2 \le k \le 6$) of Remark 4.3, verifying exact matching number $\nu(G_{8k}) = 4k$, packing numbers $\rho_3(G_{8k}, H) = k$ and $\rho_{\mathrm{m}}(G_{8k}, H) = 2k$, and explicit $\mathbb{F}_2$ kernel dimension $4k$; and it evaluates stroke-law counts (Proposition 5.1), parity-conditioned straddle sums (Theorem 5.2), super-Fibonacci band unwrapping (Theorem 5.3), block-path mean drift (Proposition 5.4), and the explicit reverse-Markov odd shift and three-regime error witnesses of Theorem 5.5 (Remark 5.7). None of the proofs in the paper depends on this script.

Module guide (`lean/DirectedReciprocity/`).

- `PositionalNest.lean`: stateless positional characterization of `HalfRate.RelCharged`, disjoint-record nesting simplification, first-appearance supply forcing, and proper record coloring across digons (Proposition 3.1).
- `StrideCycles.lean`: positional locking lemma on stride-3 hub cycles, unbounded-m piece impossibility when $s = 3$, and 2-record (even $s \ge 4$) and 3-record (all $s \ge 4$) co-supplied piece certificates on canonical stride-s cores (Proposition 3.2).
- `DigonCycle.lean`: universal chordless digon cycle dichotomy ($L \ge 4$), non-stride-3 witness extraction, and two-sided $(L+1)$-record piece expansion (Theorem 3.4).
- `PetersenHierarchy.lean`: equivalence between global `HalfRate.PieceCert` and `HalfRate.ElimCert`, canonical partition reduction and unbounded-m global impossibility on $\mathrm{GP}(5,2)$ (Petersen graph), $m \le 3$ impossibility and $m = 4$ certificate on $\mathrm{GP}(7,2)$, and $m = 3$ certificate on $\mathrm{GP}(9,2)$ (Proposition 3.5).
- `ClassS.lean`: definition of $\mathcal{S}$ and relative `ClassSOn`, condensation to an in-closed strongly connected component, minimal chordal one-way arc certificates, shortest digon cycle and $(g+2)$-record sequential hub augmentation, Theorem S* (Theorem 3.6), and the directed barrier bound Corollary B* (Corollary 3.7).
- `ClassSExamples.lean`: unbounded-m core impossibility on X_{15} (Corollary 3.3), and construction and complete cycle/peeling analysis of the 21-vertex strongly connected triangle-free witness $\mathrm{G}_{21} \in \mathcal{S}$ (Proposition 3.8).

- `TheoremSHStar.lean`: unified final count `F3_star` under (S1) alone, odd-state lemma `odd_state_star`, minimal-counterexample exclusion removing (S2), Theorem SH* in matrix, transpose, and graph forms (Theorem 4.1), and the common-neighbor-free packing invariant `rhoMatch` and Tutte–Berge deficiency bound Corollary M* (Corollary 4.2).

- `DNPStrokes.lean`: dyadic scales, stroke lengths, recursive stroke decomposition bijection, stopping-depth and parity-independence counts, and $O(m)$ recursive turn potential `evalPhiRec` (Proposition 5.1).

- `DNPStraddle.lean`: straddle reflection involution, $\mathbb{Z}$-cancellation of nested stroke potentials, m-free mod-q pair and set bounds, and exact parity-conditioned straddle count $\Pi_\kappa^2/4$ (Theorem 5.2).

- `DNPFibonacci.lean`: blind apex and valley-band error formulas on `strokePhiRec`, exact $\mathbb{Z}$-unwrapping under $\leq q/8$ bounds, super-Fibonacci step $b_0(k) \geq b_0(k+1) + b_0(k+2) + 4$, and the $2\lfloor \log_2 q \rfloor + 2$ consecutive-level bound (Theorem 5.3).

- `DNPConjectureApp.lean`: periodic block-cancellation family $B(a, K)$, forward/backward path bounds, positive mean drift, Mersenne binary-clock straddle failure (Proposition 5.4), prefix lockstep of nested V-apices, simultaneous reverse-Markov odd-shift selection, fine-scale doubling overflow, and the three-regime completeness Theorem A (Theorem 5.5).

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